UM-2 provenance

Where every ingredient, equation and number of the universe model UM-2 comes from, how it was computed, and how it was verified

Underlying theory: Patrick L. Nash Derived and verified by Claude (Anthropic)

Provenance, version 3, 2026-10-18 (version 2: 2026-10-18; version 1: 2026-10-17)

Repository once-ere/Dirac_claude, folder universe_model_2

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1 How to read this document

This document records the provenance of UM-2, a model of the Universe built on the author's 4+4-dimensional Einstein-Lovelock theory and his dirac16complex field. For every ingredient it states where it comes from, writes the ingredient out as an equation, names the computation that produced or used it, and states how it was verified. The document is self-contained: every equation it mentions is written here.

Version 2 (2026-10-18) contains three generations of quarks and leptons and three neutrino flavours with their mixing and their flavour conversion, ordered by the author on 2026-10-18; it removes the chiral tip Z portal of version 1. What changed, and why, is listed in the section on corrections (item 21); the independent review of version 2 and its fixes, among them the finding that the relic abundance of the dark matter cannot be produced by the gauge-kinetic portal within perturbative unitarity, are item 22.

Version 3 (2026-10-18) answers the author's order of 2026-10-18 to find and document an accurate approximate solution of the coupled field equations with Kohn-Sham theory for dirac16complex coupled to gravity and to every Standard-Model field: the new portal I13, the benchmark strengths of I11 and I12, the coupled solution in environments, the thermal history and the dark-matter relic, the magnitudes of the terms of the metric series, σ_8 and the Majorana alternative; its provenance is recorded in every section below and its corrections in item 23; the independent review of version 3 and its fixes are item 24.

Labels used throughout:

- **AUTHOR**: taken from the author's Mathematica notebooks or the author's written orders;
- **RECORD**: taken from the Revision record of this repository (earlier work carried out under the author's orders, with its own verification);
- **STANDARD**: textbook physics or a measured value; measured values are marked *verified* (read from the source on 2026-10-17 or 2026-10-18) or *from memory* (quoted from memory of the standard tables, to be checked against the current editions);
- **ASSUMED**: a modelling choice made for UM-2;
- **INVENTED**: a new phenomenological interaction of UM-2;
- **DERIVED**: computed and verified in UM-2;
- **ESTIMATE**: an order-of-magnitude evaluation, not a precision computation;
- **OPEN**: not resolved in this version.

Units: $\hbar = c = 1$; H is the author's constant (an inverse length); in the numerical work $H = 1$ unless a physical unit is stated.

Symbols with more than one use: k is the block variable $1/f^2$, the Lovelock order as the index of α_k and $E_{(k)}$, and the momentum of a mode along x_1, x_2, x_3 in e^{ikx_1} and h_j; $\Lambda = -10H^2$ is the eight-dimensional cosmological constant, $\Lambda = (m\,fL)^2$ in the x_0 eigenvalue problems a dimensionless eigenvalue, and $\Lambda_\star$ the scale of the effective interactions; $\xi = y/L$ is the scaled hidden coordinate (primes denote $d/d\xi$ in the methods), while the subscript t (p_t, v_t, x_t) refers to the extra times and s_h, t_h are the 4×4 blocks of the gamma matrices; ρ is an energy density, except for the electroweak parameter $\rho = \Lambda_W/(\Lambda_Z \cos^2\theta_W)$ (measured value ρ_0), and ϱ a probability density in ξ; θ_W is the angle of the gauge couplings; n is a number density, except as a level index (u_n, $h^{(1,n)}$, $n \geq 1$) and as the basis size in the Richardson formula, where p is the order; w_k are the Lovelock weights, $w = 1/2$ the weight of a fold parity, $w = \sqrt{c^2 - e^2}$ in the fermion mass condition and w an equation-of-state parameter in the supernova fits; σ is the radion in the metric expansion and, with indices, an uncertainty or a Pauli matrix; $i, j = 1, 2, 3$ label generations and neutrino mass states and $\alpha \in \{e, \mu, \tau\}$ flavours; U is the PMNS and V the CKM matrix (a matter potential carries a subscript, V_{CC}); κ (κ_ν, κ_e, κ_u, κ_d, κ_f) is a tip matrix; e is the eigenvalue $\varepsilon f L$ of a fermion, the electron and the elementary charge; μ is a bulk mass and the muon, μ_N the dark-matter-nucleon reduced mass, and $\mu = \sqrt{2} G_F n_{\nu_e}$ (no subscript) the unit of the neutrino self-interaction in the supernova analysis. The time is always x_4 (dots denote d/dx_4), and the symmetry axis of the fast-flavour analysis is x_3; $A = \dot{a}_4$ and $h = \dot{f}/f$ are the expansion rates of 3-space and of the block (h_ν, with a subscript, the comoving heating of the neutrinos); H is

the author's constant, except in the neutrino-decoupling equations, where $H = \sqrt{8\pi\rho/3}/M_{Pl}$ is the Hubble rate, and in the thermal (Boltzmann, freeze-in and right-handed-neutrino) equations of version 3, where it is the Hubble rate $H = (8\pi^3 g_\rho/90)^{1/2} T^2/M_{Pl}$ (H_0 the Hubble constant today); $F(T_1, T_2)$ is the energy-transfer kernel of the decoupling equations, F_h the decoupling factor of ΔN_{eff} and $F_{\mu\nu}$ a field strength; $G(k)$ is the Einstein-Hilbert coefficient of the block, $G(u)$ the electron-lepton-number distribution of the fast-flavour analysis with its moments J_n, I_n and Λ^μ (upper index), and G_N, G_F, G_g, G_W, G_B are couplings; $\eta = HL = 3$ is the depth of the cell and $\eta^{\mu\nu} = \mathrm{diag}(1, -1)$ the flat metric on (x_4, x_3) in the dispersion relation; α is a flavour index, the fine-structure constant and, in the fast-flavour analysis, the density ratio $n_{\bar{\nu}_e}/n_{\nu_e}$ (α_s the strong coupling; α_n, with a level index, the Yukawa strength of a Kaluza-Klein graviton). Version 3: G_V, G_A and $G = (8G_g^2 + 3G_W^2 + G_B^2)^{1/2}$ are the strengths of I11, I13 and I12 (G_A^{B-L} the strength a $B - L$ variant of I13 would need, G_λ that of the self-interaction channel, G_f the coupling of one fermion multiplet, G_i the I12 strength of the gauge group i); $\tilde{I}_{of}$, $\tilde{I}_{of}^{(b)}$, $\tilde{I}_\lambda$ are overlap integrals, s_{DF} a transition scalar density, q_f a $B - L$ charge and q_f^B a baryon number; $P_a = (1 + \gamma^{x_0})/2$, $P_b = (1 - \gamma^{x_0})/2$; J^B the baryon-number current, while J alone is the total angular momentum of a partial wave; σ is also a cross-section (σv, σ_n, σ_{SI}) and σ_8 the amplitude of the matter fluctuations; in the thermal equations $x = M/T$ with M the dark-matter mass, $Y = n/s$ with s the entropy density, but s in $\sigma(s)$, $\sqrt{s}$ and the thermal averages is the Mandelstam variable (the squared centre-of-mass energy), g_ρ, g_s the effective numbers of degrees of freedom ($g_{s,e\gamma}$ that of the photons and $e^\pm$ alone), $\delta = \frac{1}{3}\, d \ln g_s / d \ln T$, γ a rate density, K_1, K_2 modified Bessel functions of the second kind, $\beta = \sqrt{1 - 4M^2/s}$ the velocity of a dark-matter particle in the centre-of-mass frame (β_W, β_Z are tip parameters), $v = 2\beta$ the relative velocity, p the centre-of-mass momentum, N_c the number of Weyl components of a multiplet, which share its overlap (3 for a quark singlet, 6 for a quark doublet, 2 for a lepton doublet, 1 for a charged-lepton singlet), $N_\chi = N_F = 2$ the number of fold bands (the fold zero modes of Ψ_A and Ψ_B) and $\sum_{A,B}$ the sum over the two dark-matter species; c_ρ, c_3, c_p are the coefficients of the adiabatic block response, $\bar{p} = (p_0 + 3p_t)/4$ the hidden pressure and p_2 the order-α pressure of a gauge plasma; Δ_Φ is the rms Newtonian potential per logarithmic wavenumber, and $a = 3\eta/2$ in the graviton profiles (a otherwise the scale factor or, with b, a component of an orbital).

2 The orders

2.1 The order of 2026-10-17

The model answers the author's order of 2026-10-17. Its key parts, in the author's words: "Create a new model of the Universe that includes [a] gravity (Einstein-Lovelock gravity); [b] the dirac16complex quantized spinor field (which is hypothesied to represent dark matter); the dirac16complex is coupled to gravity using a non-trivial canonical spin connection; ... interactions of gravity and dirac16complex with SOME of the following Standard Model particles: [c] photons; [d] neutrinos; [e] electrons; [f] quarks (only up and down, for the protons and neutrons); [g] the 3 weak interaction gauge bosons, the W and Z 'particles'. However do NOT include the Higgs particle. You must invent/create new phenomenological interactions ... consistent with qcd and electro-weak models. You want to be able to model: CMB, CNB, BAO, so your model for the metric tensor ... must be very general, but perhaps expressible as a power series of terms of decreasing magnitude. Put each and every interaction in equation form, and explain and analyze it in documentation and provenance files; these should be both markdown and latex file types; compile all latex to pdf. NEVER refer the reader to some other file! PUT EVERY EQUATION you create and/or use IN THESE FILES. CRITICALLY IMPORTANT: Mass will be given to all non-massless particles, (except the massless particles: graviton, photon and perhaps the quanta of the dirac16complex field), by appropriate wave function dependence on the space coordinate x0 ... periodic in x0 with period $\pi/(6H)$. Localization with respect to x0 implies, using Heisenberg ... $\Delta x_0\, \Delta p_{x_0} \geq \hbar/2$ ... [2] push your new, improved, model ... along with provenance (both .md and .pdf files). [3] Compute the coupled field equations and push to repo; create concomitant documention (both .md and .pdf files) and push. [4] Compute an accurate approximate solution to the coupled field equations (use Kohn-Sham theory for the dirac16complex field, which couples to all other fields) and push ... Check and verify this repo."

Further orders of the same day: "analyze the x0 mass mechanism, and ... analyze the validity of associating dark matter (and dark energy) with dirac16complex"; "the W^+ and W^- bosons carry positive and negative electric charges. Do not forget this"; the coordinates are always the author's $\{x_0, x_1, \ldots, x_7\}$ of the notebook `Dirac_Einstein-Lovelock_4+4_Nash.nb`, never relabelled; every verified result is pushed to the repository as soon as it is obtained; no single calculation runs longer than 7 minutes and 3 seconds (423 s) (this limit was lifted

on 2026-10-18).

2.2 The orders of 2026-10-18: three neutrino flavours

On 2026-10-18 the author stated that neutrino flavour and flavour conversion are critically important in this project and asked that the paper of Gogilashvili and Tamborra on neutrino flavour conversion in core-collapse supernovae (his file "Neutrino Flavor Conversion.pdf", arXiv:2605.16504; not committed, it is a third-party paper) be read, understood and remembered (paraphrase). He then ordered, in his words: "correct universe model 2 to include three neutrino flavors with mixing and flavor conversion. this is your first and only priority!! ... do correct universe model 2 to include three neutrino flavors with mixing and flavor conversion NOW." Two ways were put to him: (A) three full generations of quarks and leptons, anomaly free generation by generation; (B) three lepton generations with the single quark generation of version 1, whose gauge anomalies would have to be cancelled by an added Wess-Zumino term. His answer: "NOT (B). USE (A)!!!". Version 2 implements (A). A flavour is defined by the charged lepton produced with the neutrino in a charged-current process, so three neutrino flavours require the muon and the tau; with them, anomaly freedom requires the second and third quark generations (a single quark generation would leave $\mathcal{A}_{SU(2)^2U(1)} = -1$ and $\mathcal{A}_{U(1)^3} = 3/2$; DERIVED).

2.3 The order of 2026-10-18 for version 3

After the three-flavour correction the author ordered, in his words (excerpts): "find and document (with provenance) an accurate approximate solution for your new model universe, i.e., of the coupled field equations, with Kohn-Sham theory for the dirac16complex field coupled to gravity and to all of the Standard-Model fields; coupled fields include gravity (Einstein-Lovelock gravity, which couples to all fields), the dirac16complex quantized fermion spinor field (which couples to all other fields), the three-flavour version of left-handed neutrinos, the usual left-handed electrons, the usual singlet right-handed electron, three quark generations, gluons, photons, the 3 weak interaction gauge bosons, the Wˆ(+), Wˆ(-), and Z 'particles'. However do NOT include the Higgs particle. You must invent/create and document new phenomenological interactions for these standard model particles that should be consistent with qcd and electro-weak models. You want to be able to model: CMB ..., CNB ..., BAO ..., so your model for the metric tensor of the gravitation field must be very general, but perhaps expressible as a power series of terms of decreasing magnitude. CRITICALLY IMPORTANT: Mass will be given to all non-massless particles ... by appropriate wave function dependence on the space coordinate x0 ..., which must be periodic in x0 with period $\pi/(6H)$. ... Put each and every interaction in equation form, and explain and analyze it in documentation and provenance files; these should be both markdown and latex file types; compile all latex to pdf. NEVER refer the reader to some other file! PUT EVERY EQUATION you create and/or use IN THESE FILES. In your documentations and provenances i want you to analyze the x0 mass mechanism, and to analyze the validity of associating dark matter (and dark energy) with dirac16complex. [2] push your new, improved, model of the Universe to the repo, along with provenance (both .md and .pdf files). [3] Compute the coupled field equations and push to repo ... [4] Compute an accurate approximate solution to the coupled field equations (use Kohn-Sham theory for the dirac16complex field, which couples to all other fields) and push to repo ... Check and verify this repo. DO NOT TAKE SHORTCUTS. ... [5] Continue and Complete and push all stages that are still open". Version 3 keeps the three generations of version 2 (with right-handed neutrinos for the Dirac masses of the x_0 mechanism; the Majorana alternative without them is analysed) and adds the items listed above.

3 Who did what

- The 4+4 framework, the coordinates, the primordial metric of the notebooks, the Einstein-Lovelock action with its weights and Λ, the dirac16complex field and its $Cl(4,4)$ gamma matrices, the mass parameter M, the idea that masses come from the x_0 dependence of wave functions periodic with period $\pi/(6H)$, the dark-matter hypothesis and the demand for three neutrino flavours with mixing and flavour conversion are the AUTHOR's (Patrick L. Nash).
- The corrected primordial metric (version 2), the regular tip cutoff, the Dirac-type Lagrangian with the canonical spin connection, the 2×2 block reduction of the Kohn-Sham problem and its conventions are RECORD items: produced earlier in this repository under the author's orders and verified there.
- Everything labelled ASSUMED, INVENTED or DERIVED below, all the code of UM-2 and all its documents are by Claude (Anthropic), working in this repository on 2026-10-17 (version 1) and 2026-10-18 (versions 2 and 3).

4 Ingredients from the author (AUTHOR)

4.1 Coordinates and signature

The author's coordinates (notebook `Dirac_Einstein-Lovelock_4+4_Nash.nb`, `X = {x0[], x1[], ..., x7[]}`) are

$$X = \{x_0, x_1, x_2, x_3, x_4, x_5, x_6, x_7\}:$$

x_0 a hidden space-like direction, x_1, x_2, x_3 ordinary space, x_4 the time, x_5, x_6, x_7 three extra time-like directions. The orthonormal frame metric is

$$\eta_{ab} = \mathrm{diag}(+1,+1,+1,+1,-1,-1,-1,-1), \qquad a,b = 0,\ldots,7 \text{ aligned with } x_0,\ldots,x_7.$$

The hidden coordinate enters through $z = 6Hx_0$; one period of x_0 is $\ell_0 = \pi/(6H)$.

4.2 The primordial metric of the notebooks

The notebook `WaveFunctionUniverse-4+4-Einstein-Lovelock.nb` (the author's local notebook) gives the primordial metric

$$g^{\mathrm{nb}} = \mathrm{diag}\big(\cot^2 z,\ e^{2a_4}(\sin^2 z)^{1/6},\ e^{2a_4}(\sin^2 z)^{1/6},\ e^{2a_4}(\sin^2 z)^{1/6},$$
$$-1,\ -e^{-2a_4}(\sin^2 z)^{1/6},\ -e^{-2a_4}(\sin^2 z)^{1/6},\ -e^{-2a_4}(\sin^2 z)^{1/6}\big),$$

with vielbein $\mathrm{diag}(\cot z, e^{a_4}(\sin^2 z)^{1/12}\times 3, 1, e^{-a_4}(\sin^2 z)^{1/12}\times 3)$ and $a_4 = a_4(x_4)$. By the author's orders of 2026-10-14 a second function $b_4(x_4) > 0$ multiplies the extra-time components ("b4 is a function of x4; b4 is NEVER 1"), and $(\sin z)^{1/3}$ means $(\sin^2 z)^{1/6}$, the positive branch, for every real z.

4.3 The Einstein-Lovelock action, weights and cosmological term

The notebook writes the Einstein-Lovelock vacuum equations (after Lovelock and Rund) as

$$0 = -\Lambda^{\mathrm{nb}} + H^{-2} w_1\,\mathrm{Lovelock1} + H^{-4} w_2\,\mathrm{Lovelock2} + H^{-6} w_3\,\mathrm{Lovelock3},$$

where Lovelock$k$ denotes the notebook's Lovelock tensor of order k, the tensor ${E_{(k)}}^M{}_N$ below, with $(w_1, w_2, w_3, \Lambda^{\mathrm{nb}}) = (1, 1/32, 1/1152, 10)$: in the notebook `Dirac_Einstein-Lovelock_4+4_Nash.nb` (In[133] to In[136]) the author sets $w_1 = 1$, $w_2 = 1/2^5$ and solves the vacuum equations on the history $a_4 = Hx_4$, $b_4 = 1$ of the notebook metric for w_3 and Λ^{nb} (In[134]); In[136] factors $1152 = 2^7\,3^2$; the Revision specification records the solution $w_3 = 1/1152$, $\Lambda^{\mathrm{nb}} = 10$. Multiplying by H^2 and writing the equations as $\sum_k \alpha_k {E_{(k)}}^M{}_N + \Lambda\,\delta^M_N = \kappa T^M{}_N$ gives the UM-2 form

$$\alpha_k = w_k H^{2-2k}, \qquad w = \Big(1, \frac{1}{32}, \frac{1}{1152}\Big), \qquad \Lambda = -\Lambda^{\mathrm{nb}} H^2 = -10H^2,$$

where

$${E_{(k)}}^M{}_N = -\frac{1}{2^{k+1}}\,\delta^{M A_1 B_1 \cdots A_k B_k}_{N C_1 D_1 \cdots C_k D_k}\, R^{C_1 D_1}{}_{A_1 B_1} \cdots R^{C_k D_k}{}_{A_k B_k},$$

$$\mathcal{L}_{(k)} = \frac{1}{2^k}\,\delta^{A_1 B_1 \cdots A_k B_k}_{C_1 D_1 \cdots C_k D_k}\, R^{C_1 D_1}{}_{A_1 B_1} \cdots R^{C_k D_k}{}_{A_k B_k},$$

$\delta^{\cdots}_{\cdots}$ being the generalised Kronecker delta (the determinant of ordinary deltas); $E_{(1)}$ is the Einstein tensor, $\mathcal{L}_{(1)} = R$, $\mathcal{L}_{(2)} = R^2 - 4R_{AB}R^{AB} + R_{ABCD}R^{ABCD}$. That the UM-2 normalisation reproduces the record's equations (which were checked against the notebook) is verified exactly (script `um2_geometry.py`, checks `record_equation`). UM-2 keeps the author's weights and Λ unchanged; an earlier model of this repository (UM-1) had changed w_3 and Λ.

4.4 The gamma matrices of $Cl(4,4)$

The author's notebooks (*Dirac_Einstein-Lovelock_4+4_Nash.nb* and *WaveFunctionUniverse-4+4-Einstein-Lovelock.nb*) build real 16×16 matrices from ($h = 1,2,3$; $p,q = 1,\ldots,4$; ϵ the Levi-Civita symbol)

$$Q_a(h,p,q) = \epsilon_{hpq4}, \qquad Q_b(h,p,q) = \delta_{p4}\delta_{qh} - \delta_{ph}\delta_{q4}, \qquad s_h = Q_a - Q_b, \quad t_h = Q_a + Q_b,$$

$$\zeta_0 = I_8, \quad \zeta_{7-h} = \begin{pmatrix} 0 & t_h \\ -t_h & 0 \end{pmatrix}, \quad \zeta_h = \begin{pmatrix} 0 & s_h \\ s_h & 0 \end{pmatrix}, \quad \zeta_7 = \zeta_1\zeta_2\zeta_3\zeta_4\zeta_5\zeta_6, \quad \sigma = \begin{pmatrix} 0 & I_4 \\ I_4 & 0 \end{pmatrix},$$

$$\tau_0 = I_8, \quad \tau_A = \zeta_A, \quad \bar\tau_A = \sigma\,\tau_A^T\,\sigma, \qquad \gamma^{x_A} \equiv T16[A] = \begin{pmatrix} 0 & \bar\tau_A \\ \tau_A & 0 \end{pmatrix} \quad (A = 0, \dots, 7).$$

They satisfy $\{\gamma^a, \gamma^b\} = 2\eta^{ab} I_{16}$ and the notebook identities $\frac{1}{2}(\tau_A\bar\tau_B + \tau_B\bar\tau_A) = \eta_{AB} I_8$, $\sigma = \tau_1\tau_2\tau_3 = \tau_4\tau_5\tau_6\tau_7$ (notebook Out[322] to Out[324]). Derived objects used by UM-2:

$$C = \gamma^{x_0}\gamma^{x_1}\gamma^{x_2}\gamma^{x_3} = \begin{pmatrix} -\sigma & 0 \\ 0 & \sigma \end{pmatrix}, \qquad \Gamma = \gamma^{x_0}\cdots\gamma^{x_7} = \mathrm{diag}(-I_8, I_8), \qquad B = -iC\gamma^{x_4}, \qquad S^{ab} = \tfrac{1}{4}[\gamma^a, \gamma^b].$$

4.5 The mass parameter, the x_0 mass idea and the dark-matter hypothesis

The dirac16complex mass is written $m = 3HM$ in the author's notebooks. The order requires every non-massless particle to get its mass from the dependence of its wave function on x_0, periodic with period $\pi/(6H)$, with the uncertainty relation $\Delta x_0\,\Delta p_{x_0} \geq \hbar/2$ as the physical reason, and states the hypothesis that the quanta of dirac16complex represent dark matter (and asks for the validity of a dark-energy association to be analysed).

5 Ingredients from the Revision record (RECORD)

5.1 The corrected primordial metric (version 2)

The author's order of 2026-10-16: "a diagonal form for the metric of the primordial gravitational field is only self-consistent if all off-diagonal terms of all source terms in the Einstein field equations ARE ZERO. This is not the case here. I task you to: [1] find the simplest modification for the current metric of the primordial gravitational field that is probably self-consistent with the Einstein field equations". The metric of 2026-10-14 (the notebook metric with b_4) has the off-diagonal component $E_{(k)}{}^{x_4}{}_{x_0} \propto \cot z\,\dot b_4/b_4$ at every Lovelock order. In the family (with $g_{44} = -1$)

$$g_{00} = b_4^{2p} e^{2qa_4}\cot^2 z, \qquad g_{ii} = e^{2a_4}(\sin^2 z)^{3\alpha}, \qquad g_{tt} = -e^{-2a_4} b_4 (\sin^2 z)^{3\beta}$$

the mixed curvature is proportional to $\alpha\big((q-1)\dot a_4 + p\,\dot b_4/b_4\big)$ (3-space) and $\beta\big((q+1)\dot a_4 + (p-\frac{1}{2})\dot b_4/b_4\big)$ (extra times). Keeping the warp of the extra times ($\beta = 1/6$) forces $\alpha = 0$, $q = -1$, $p = 1/2$: the record's current metric, used by UM-2,

$$\bar g = \mathrm{diag}\big(e^{-2a_4} b_4\cot^2 z,\ e^{2a_4},\ e^{2a_4},\ e^{2a_4},\ -1,\ -e^{-2a_4} b_4 (\sin^2 z)^{1/6},\ -e^{-2a_4} b_4 (\sin^2 z)^{1/6},\ -e^{-2a_4} b_4 (\sin^2 z)^{1/6}\big).$$

Every off-diagonal component of $E_{(1)}, E_{(2)}, E_{(3)}$ vanishes identically for all $a_4(x_4)$, $b_4(x_4) > 0$ and every real z (record: sympy 14/14, Wolfram 13/13; re-derived in UM-2, `um2_geometry.py`). The alternative K of the record ($g_{00} = e^{2a_4}\cot^2 z$, $g_{ii} = e^{2a_4}(\sin^2 z)^{1/6}$, $g_{tt} = -e^{-2a_4} b_4$) is also diagonal at every order; the record leaves the choice to the author, and UM-2 uses version 2.

5.2 The hidden coordinate, the fold and the regular tip

On every interval with $\sin z \neq 0$,

$$y = \frac{\ln|\sin z|}{6H} \in (-\infty, 0], \qquad dy = \cot z\,dx_0, \qquad f = e^{-a_4} b_4^{1/2},$$

$$ds^2 = e^{2a_4} d\vec{x}^{\,2} - dx_4^2 + f^2\big[dy^2 - e^{2Hy}(dx_5^2 + dx_6^2 + dx_7^2)\big].$$

The fold is $z = \pi/2$ ($y = 0$); the tips are $z \to 0^+$ and $z \to \pi^-$ ($y \to -\infty$). The record replaces the tip by a regular cutoff at $y = -L$ with

$$\eta \equiv HL = 3,$$

a chosen cutoff (the extended levels depend on it like those of a box of proper length fL). UM-2 keeps $\eta = 3$ (ASSUMED).

5.3 The dirac16complex Lagrangian and the canonical spin connection

The specification of the record (under the author's orders) fixes the Dirac-type Lagrangian

$$\mathcal{L}_\Psi = \sqrt{|g|}\Big[\tfrac{1}{2}\big(\bar\Psi\gamma^M D_M\Psi - (D_M\bar\Psi)\gamma^M\Psi\big) - m\,\bar\Psi\Psi - U(S)\Big], \qquad S=\bar\Psi\Psi, \quad U(S)=\frac{\lambda}{2}S^2, \quad \bar\Psi=\Psi^\dagger C,$$

$$D_M\Psi = \partial_M\Psi + \Omega_M\Psi, \quad \Omega_M = \tfrac{1}{2}\omega_{Mab}S^{ab}, \quad \omega_M{}^a{}_b = e^a{}_N\big(\partial_M e_b{}^N + \Gamma^N{}_{ML}e_b{}^L\big), \quad \gamma^M = e_a{}^M\gamma^a,$$

the canonical (vielbein-postulate, torsion-free) spin connection. The record chose the Dirac-type form because the notebook's real Majorana-type Lagrangian is a total derivative for anticommuting real fields (it has no field equations), and it uses the canonical connection rather than the notebook's contraction of the mixed $\omega_M{}^a{}_b$, which lacks a metric factor. The field equation is $\gamma^M D_M\Psi = (m + U'(S))\Psi$.

5.4 Canonical quantisation and the Krein metric

With x_4 as the evolution time, $\{\Psi_r(x), \Psi_s^\dagger(x')\} = B_{rs}\,\delta^7(x-x')/\sqrt{|g|}$ (r,s spinor indices) on a slice $x_4 = \text{const}$; B has eigenvalues $+1$ (8 times) and -1 (8 times), so the inner product is indefinite (a Krein space). The record works in the good sector (no extra-time momentum), counts the positive-energy branch of every block as particles, and evaluates expectation values as $\langle\Psi^\dagger M\Psi\rangle = u^\dagger BMu$.

5.5 The Kohn-Sham reduction

For the corrected metric the record reduces the 16-component equation, with $\Psi = e^{ikx_1}\,e^{-\frac{3}{2}Hy}\,e^{a_4/2}b_4^{-1}\,V_\beta\,\chi(y,x_4)$ (k the momentum along x_1), to the 2×2 block Hamiltonians

$$h_j = j\Big[-\frac{i}{f}\sigma_1\partial_y + M(y)\,\sigma_2 + e^{-a_4}k\,\sigma_3\Big] + v(y),$$

in the blocks (j, s_2, s_3) of the commuting set $J=\gamma^{x_0}\gamma^{x_1}\gamma^{x_4}$, $K_1=\gamma^{x_2}\gamma^{x_3}$, $K_2=\gamma^{x_5}\gamma^{x_6}$ (eigenvalues j, is_2, is_3), with the contact-interaction mean fields of the record's Hartree-plus-exchange treatment of the contact interaction

$$e_{int} = \frac{15}{32}\lambda S^2 - \frac{1}{32}\lambda n^2, \qquad M_{eff} = m + \frac{15}{16}\lambda S, \qquad v = -\frac{\lambda}{16}n,$$

the fold conditions $\theta\in\{0,\pi\}$ (both parities filled, weight $w=1/2$ each), the regular tip, and the densities carrying $P = e^{-3Hy}/\mathrm{Vol}_7$, with $\mathrm{Vol}_7 = (2\pi/\Delta k)^3 v_t f^4$ the normalisation volume of a cell (Δk the spacing of the momentum lattice, v_t the coordinate volume of the extra times, $a_4=0$; the proper seven-volume of a cell is $2\,\mathrm{Vol}_7(1-e^{-3\eta})/(3H)$). The record's mass is $m = 3HM$ with $M=1/3$, i.e. $m=H$; UM-2 keeps it (ASSUMED). Two reference states of the record (at $f=1$, $a_4=0$: $N=8$ with $\lambda=157.7$, and $N=592$ with $\lambda=2.131$) serve as validation targets of the independent UM-2 solver.

5.6 Earlier work used only for comparison

The first universe model of this repository (UM-1, folder `Revision/universe_model`) is not an input of UM-2. UM-2 differs from it in keeping the author's weights and Λ, in the electroweak sector, in the number of generations and in the dark-matter candidate. Version 1 of UM-2 (2026-10-17, one generation) is superseded by version 2; its differences are listed in item 21 of the corrections.

6 Standard physics and measured values (STANDARD)

Every measured value enters UM-2 in one of three roles, never as a derived result: as the target of a fit (the $W^\pm$ and Z masses with ρ_0, the twelve fermion masses, the two neutrino splittings with the chosen m_1, and $\Omega_c h^2$ for the freeze-out requirement), as a direct input (the angles and phases of the PMNS and CKM matrices, G_F, the $\overline{MS}$ value of $\sin^2\theta_W$ in the Z widths, α_s, α, T_0, the Planck 2018 parameters of the cosmology, the effective couplings and kernel factors of the decoupling equations, the matter densities and the supernova parameters), or as a comparison (the Z widths, N_ν, N_{eff}, the BAO, the explosion statistics, the supernova dark-energy fits, the LZ limit).

input	value used	source	status
$W^{\pm}$ mass	80.3625 ± 0.0077 GeV	PDG 2026, OUR EVALUATION (pdgLive S043M)	verified
Z mass	91.1879 ± 0.0020 GeV	PDG 2026, OUR AVERAGE (pdgLive S044M)	verified
$\sin^2\theta_W$ ($\overline{MS}$, M_Z)	0.23122 ± 0.00004	PDG electroweak review	from memory
electroweak ρ parameter	$\rho_0 - 1 = 0.00038 \pm 0.00020$	PDG electroweak review (global fit)	from memory
charged-lepton masses	$m_e = 0.51099895000$ MeV, $m_\mu = 105.6583755$ MeV, $m_\tau = 1776.93 \pm 0.09$ MeV	PDG 2025 summary tables	verified
light-quark masses ($\overline{MS}$, 2 GeV)	$m_u = 2.16 \pm 0.07$ MeV, $m_d = 4.70 \pm 0.07$ MeV, $m_s = 93.5 \pm 0.8$ MeV	PDG 2025 summary tables	verified
heavy-quark masses	$m_c(m_c) = 1.2730 \pm 0.0046$ GeV, $m_b(m_b) = 4.183 \pm 0.007$ GeV, $m_t = 172.56 \pm 0.31$ GeV (direct)	PDG 2025 summary tables	verified
neutrino oscillation parameters (normal ordering)	$\sin^2\theta_{12} = 0.308$, $\sin^2\theta_{23} = 0.470$, $\sin^2\theta_{13} = 0.02215$, $\delta = 212°$, $\Delta m^2_{21} = 7.49 \times 10^{-5}$ eV^2, $\Delta m^2_{31} = 2.513 \times 10^{-3}$ eV^2; inverted ordering $\Delta\chi^2 = 6.1$; $J_{max} = 0.0333 \pm 0.0007$	NuFIT 6.0, arXiv:2410.05380 (JHEP 12 (2024) 216), Table 1, IC24 with SK atmospheric data	verified
lightest neutrino mass	$m_1 = 1.000$ meV	chosen	ASSUMED
sum of neutrino masses	$\sum m_\nu < 0.064$ eV (95 %, ΛCDM)	DESI DR2 Results II, arXiv:2503.14738	verified
CKM parameters	$s_{12} = 0.22501$, $s_{13} = 0.003732$, $s_{23} = 0.04183$, $\delta = 1.147$ rad; $J = 3.12 \times 10^{-5}$	PDG 2024 review "CKM quark-mixing matrix", global fit	verified
strong coupling	$\alpha_s(m_Z) = 0.1180$	PDG world average	from memory
Fermi constant, electroweak scale	$G_F = 1.1663788 \times 10^{-5}$ GeV^{-2}, $v = 246.22$ GeV	PDG	from memory
fine-structure constant	$\alpha^{-1}(0) = 137.035999$	CODATA	from memory
number of light neutrinos	$N_\nu = 2.9963 \pm 0.0074$	PDG 2025 (Janot and Jadach, Phys. Lett. B803 (2020) 135319)	verified
(superseded, version 1)	$N_\nu = 2.9840 \pm 0.0082$	LEP/SLD, Phys. Rept. 427 (2006) 257	verified
Z width and branching fractions	$\Gamma_Z = 2495.5 \pm 2.3$ MeV; invisible 20.000 %, hadrons 69.911 %, $\ell^+\ell^-$ 3.3658 %	PDG 2025 Z listing	verified
nucleon masses	$m_p = 938.272$ MeV, $m_n = 939.565$ MeV	PDG	from memory

input	value used	source	status
$\hbar c$, k_B, N_A	$1.973269804 \times 10^{-16}$ GeV m, $8.617333262 \times 10^{-5}$ eV/K, $6.02214076 \times 10^{23}$	CODATA, SI	from memory
CMB temperature	$T_0 = 2.7255$ K	Fixsen 2009	from memory
N_{eff} in the Standard Model	3.0432 ± 0.0002 (three flavours, oscillations, NLO)	Cielo, Escudero, Mangano, Pisanti, arXiv:2306.05460	verified
N_{eff} by the method of temperatures	3.000, 3.011, 3.010, 3.039, 3.049, 3.048 (Table 1); couplings $g_{eL} = 0.727$, $g_{eR} = 0.233$, $g_{\mu L} = -0.273$, $g_{\mu R} = 0.233$; $f_a = 0.884$, $f_s = 0.829$	Escudero, arXiv:2001.04466	verified (2026-10-18)
finite-temperature QED pressure	P_2, P_3 of the $e^{\pm}\gamma$ plasma (formulas below)	Bennett et al., arXiv:1911.04504	from memory; massless limits reproduced to 10^{-16}
N_{eff} measured	2.99 ± 0.17 (Planck 2018 + BAO); 2.86 ± 0.13 (ACT DR6)	Planck 2018 VI, A&A 641 (2020) A6; Calabrese et al., arXiv:2503.14454	verified
Planck 2018 base parameters	$\omega_b = 0.02237$, $\omega_c = 0.1200$, $100\,\theta_{MC} = 1.04092$, $\tau_{reio} = 0.0544$, $\ln(10^{10} A_s) = 3.044$, $n_s = 0.9649$	Planck 2018 results VI, Table 2	from memory
BAO	DESI DR2: 13 measurements of D_V/r_d, D_M/r_d, D_H/r_d (table below)	DESI collaboration, arXiv:2503.14738	values verified (2026-10-17) against Table 2 of arXiv:2603.26861; correlations from memory
dark-matter density	$\Omega_c h^2 = 0.120$	Planck 2018	from memory
Planck mass (freeze-out)	$M_{Pl} = G_N^{-1/2} = 1.22091 \times 10^{19}$ GeV		from memory
reduced Planck mass (radion decay)	$\bar{M}_{Pl} = (8\pi G_N)^{-1/2} = 2.435 \times 10^{18}$ GeV		from memory
angle of the gauge couplings	$\cos^2\theta_W = (m_W^2/m_Z^2)/\rho_0 = 0.776368$ (used for $\beta_Z = \beta_W/\cos^2\theta_W$ and in $\rho = \Lambda_W/(\Lambda_Z \cos^2\theta_W)$); the on-shell value $m_W^2/m_Z^2 = 0.776663$ for comparison	computed from the $W^{\pm}$, Z masses and ρ_0 above	computed
solar electron density	$n_e = 245\,N_A\,e^{-10.54\,r/R_\odot}$ cm^{-3} outside the core, at most $100\,N_A$ cm^{-3} (exponential fit)	after Bahcall	ASSUMED profile
matter of the Earth	$\rho = 2.6$ (T2K), 2.84 (NOvA), 2.848 (DUNE), 2.8 g/cm^3 (resonance); $Y_e = 0.5$	representative constant densities	ASSUMED

input	value used	source	status
supernova envelope and neutrinospheres	$\rho(r) = 10^4\,(10^9\,\mathrm{cm}/r)^3$ g/cm^3, $Y_e = 0.5$; $L = 6, 6, 4 \times 10^{52}$ erg/s, $\langle E\rangle = 11, 14, 15.5$ MeV (ν_e, $\bar\nu_e$, ν_x); $n_{\nu_e} = 10^{32}$ to 10^{33} cm^{-3}	representative values	ASSUMED
supernova explosion statistics with flavour conversion	failed fractions 25.6 % (none), 50.8, 77.4, 93.3, 96.4 % ($\rho_c = 10^9$ to 10^{13} g/cm^3); Salpeter-weighted 27.0, 47.6, 61.9, 73.9, 88.3 %	Gogilashvili and Tamborra, arXiv:2605.16504 (the author's file)	verified (read 2026-10-18)
fast flavour instability criterion	instability if and only if the ELN distribution has a crossing	Morinaga, Phys. Rev. D 105 (2022) L101301	from memory
direct-detection limit	LZ: 2.2×10^{-48} cm^2 at 40 GeV, scaled linearly with the mass	LZ 2024	from memory; the extrapolation is an ESTIMATE
nucleon gluon content, strong coupling	$f_{TG} = 0.92$, $\alpha_s = 0.3$ (low scale)	standard	from memory; ESTIMATE
dark-energy density	$\rho_\Lambda \approx 2.5 \times 10^{-47}$ GeV^4		from memory
gluon condensate (version 3)	$\langle \frac{\alpha_s}{\pi} G^\alpha G^\alpha \rangle_0 = 0.012$ GeV^4	Shifman, Vainshtein, Zakharov	from memory
QCD interaction measure (version 3)	$(\rho - 3p)/T^4$ peaks at about 4 near 200 MeV (lattice)	lattice QCD (HotQCD and Wuppertal-Budapest)	from memory; the shape used is an ESTIMATE
nucleon sigma terms (version 3)	$\sigma_{\pi N} = 45$ MeV, $\sigma_s = 40$ MeV	standard	from memory; ESTIMATE
electromagnetic coupling at m_Z (version 3)	$\alpha(m_Z) = 1/127.95$	PDG	from memory
baryon-to-photon ratio (version 3)	$\eta_b = 6.1 \times 10^{-10}$	Planck 2018, BBN	from memory
neutron-proton mass difference (version 3)	$Q = 1.293$ MeV	PDG	from memory
hadron resonance gas (version 3)	π (0.138, 3), K (0.4956, 4), η (0.5479, 1), ρ (0.7753, 9), ω (0.7827, 3), K^* (0.8917, 12), N (0.9389, 8), η' (0.9578, 1), ϕ (1.0195, 3), Λ (1.1157, 4), Σ (1.1932, 12), Δ (1.232, 32) (mass in GeV, states)	PDG	from memory
QCD crossover (version 3)	$T_c = 155$ MeV, width 15 MeV (smooth switch)	lattice QCD	from memory; the switch is ASSUMED
σ_8 (version 3)	0.811	Planck 2018	from memory
tensor-to-scalar ratio (version 3)	$r < 0.036$ (95 %)	BICEP/Keck 2021	from memory

input	value used	source	status
astrophysical bodies (version 3)	$G_N = 6.67430 \times 10^{-11}$ $\mathrm{m^3 kg^{-1} s^{-2}}$; Sun 1.98892×10^{30} kg, 6.957×10^8 m; Earth 5.9722×10^{24} kg, 6.371×10^6 m; white dwarf 0.6 solar masses, 8700 km; neutron star 1.4 solar masses, 12 km, core density 0.3 $\mathrm{fm^{-3}}$ with 10 % protons; Earth mean density 5.51 $\mathrm{g/cm^3}$; solar core 150 $\mathrm{g/cm^3}$; white dwarf 10^6 $\mathrm{g/cm^3}$	standard	from memory; representative
supernova dark-energy fits	constant $w = -0.764$; $(w_0, w_a) = (-0.861, -0.60)$	the author's private file (not in the repository)	AUTHOR

The DESI DR2 BAO measurements used (z; D_V/r_d; D_M/r_d; D_H/r_d; correlation r_{MH} of D_M and D_H):

tracer	z	D_V/r_d	D_M/r_d	D_H/r_d	r_{MH}
BGS	0.295	7.942 ± 0.075			
LRG1	0.510		13.588 ± 0.167	21.863 ± 0.425	-0.459
LRG2	0.706		17.351 ± 0.177	19.455 ± 0.330	-0.404
LRG3+ELG1	0.934		21.576 ± 0.152	17.641 ± 0.193	-0.416
ELG2	1.321		27.601 ± 0.318	14.176 ± 0.221	-0.434
QSO	1.484		30.512 ± 0.760	12.817 ± 0.516	-0.500
Lyα	2.330		38.988 ± 0.531	8.632 ± 0.101	-0.431

with

$$\chi^2 = \sum_i (d_i - d_i^{th})^T C_i^{-1} (d_i - d_i^{th}), \qquad C_i = \begin{pmatrix} \sigma_M^2 & r_{MH}\sigma_M\sigma_H \\ r_{MH}\sigma_M\sigma_H & \sigma_H^2 \end{pmatrix}$$

(d_i the measured and d_i^{th} the computed pair of D_M/r_d, D_H/r_d) for the anisotropic tracers and $(d - d^{th})^2/\sigma_V^2$ for BGS, where $D_M(z) = \int_0^z c\,dz'/H(z')$ (flat), $D_H = c/H(z)$, $D_V = (z D_M^2 D_H)^{1/3}$.

Standard theory used: the Standard-Model gauge theory $SU(3)_c \times SU(2)_L \times U(1)_Y$ with three generations; QCD confinement of the quarks into hadrons (protons and neutrons); the PMNS and CKM matrices in the standard parametrisation

$$U = \begin{pmatrix} c_{12}c_{13} & s_{12}c_{13} & s_{13}e^{-i\delta} \\ -s_{12}c_{23} - c_{12}s_{23}s_{13}e^{i\delta} & c_{12}c_{23} - s_{12}s_{23}s_{13}e^{i\delta} & s_{23}c_{13} \\ s_{12}s_{23} - c_{12}c_{23}s_{13}e^{i\delta} & -c_{12}s_{23} - s_{12}c_{23}s_{13}e^{i\delta} & c_{23}c_{13} \end{pmatrix}$$

($s_{ij} = \sin\theta_{ij}$, $c_{ij} = \cos\theta_{ij}$; the CKM matrix V has the same form) with the Jarlskog invariant $J = c_{12}s_{12}c_{23}s_{23}c_{13}^2 s_{13}\sin\delta$; the flavour evolution of the amplitudes $\psi = (\psi_e, \psi_\mu, \psi_\tau)$ of a neutrino of energy E along its path, $i\,d\psi/dx_4 = [U\,\mathrm{diag}(m_1^2, m_2^2, m_3^2)U^\dagger/(2E) + \mathrm{diag}(V_{CC}, 0, 0)]\psi$ with $V_{CC} = \sqrt{2}G_F n_e$ (Wolfenstein, Mikheyev and Smirnov; antineutrinos $U \to U^*$, $V_{CC} \to -V_{CC}$); the tree-level Z partial width $\Gamma(Z \to f\bar{f}) = N_c G_F m_Z^3 (g_V^2 + g_A^2)/(6\sqrt{2}\pi)$, $g_V = T_3 - 2Q\sin^2\theta_W$ (with the $\overline{MS}$ value $\sin^2\theta_W = 0.23122$), $g_A = T_3$; the radiative decay $\mathrm{BR}(\mu \to e\gamma) = \frac{3\alpha}{32\pi}|\sum_i U_{\mu i}^* U_{ei} m_i^2/m_W^2|^2$ for Dirac neutrinos; the freeze-out of a relic (Kolb and Turner); the trace-anomaly nucleon matrix element; the Einstein-Boltzmann equations of linear cosmological perturbations (solved by CAMB); the linearised equations of a dense neutrino gas (fast flavour

conversion); Higgsless electroweak symmetry breaking by a non-linear sigma field and fermion localisation in an extra dimension (concepts only, no numbers).

7 Ingredients new in UM-2 (ASSUMED and INVENTED)

7.1 The metric as a power series (ASSUMED)

$$g_{MN} = \bar{g}_{MN} + \epsilon\, h^{(1)}_{MN} + \epsilon^2 h^{(2)}_{MN} + \ldots, \qquad \epsilon \sim 10^{-5} \text{ (the CMB anisotropy)},$$
$$h^{(1)}_{MN}(x^\mu, x_0) = \sum_{n\geq 0} h^{(1,n)}_{MN}(x^\mu)\, u_n(x_0),$$

with u_n the eigenfunctions of the x_0 operator; the flat $n = 0$ term in the Newtonian gauge,

$$ds^2 = e^{2a_4}\big[(1 - 2\epsilon\Psi_N)\delta_{ij} + \epsilon h^{TT}_{ij}\big]dx_i dx_j - (1 + 2\epsilon\Phi)dx_4^2 + f^2(1 + 2\epsilon\sigma)\big[dy^2 - e^{2Hy}d\vec{x}_t^{\,2}\big],$$

carries the CMB, CNB and BAO physics; the radion σ and the $n \geq 1$ Kaluza-Klein modes are massive.

7.2 The block stabiliser (ASSUMED; its origin is OPEN)

$$S_{stab} = -\int d^8x\sqrt{|g|}\;\rho_b(f), \qquad \kappa\rho_b(f) = \kappa\rho_b^*\Big(\frac{f_*}{f}\Big)^8,$$

$$T^M{}_N[\text{stab}] = \text{diag}(p_0, p_3, p_3, p_3, -\rho_b, p_t, p_t, p_t), \qquad p_3 = -\rho_b, \qquad p_0 = p_t = \rho_b.$$

It is conserved ($\dot{\rho}_b = -3A(\rho_b + p_3) - 4h(\rho_b + p_0) = -8h\rho_b$, so $\rho_b \propto f^{-8}$), four-dimensionally Lorentz invariant ($p_3 = -\rho_b$) and isotropic in the block ($p_0 = p_t$). It was introduced because the author's weights and Λ admit no static block with attractive four-dimensional gravity without a source: the only sourceless static block with $A = 0$ is $k_0 = 1/(4w_2) = 8$, where the coefficient of the four-dimensional Einstein-Hilbert term is $G(k_0) = 1 - 24w_2k_0 + 72w_3k_0^2 = -1$ (repulsive), with $\Lambda_0 = -24H^2 \neq -10H^2$ (`um2_background.py`). The family $p_0 = p_t = \nu\rho_b$, $\rho_b \propto f^{-4(1+\nu)}$, was also computed at $\nu = 1/4, 1/2, 3/4, 1, 5/4, 3/2, 2, 3$.

7.3 The Standard-Model fields in eight dimensions (ASSUMED structure; boundary conditions INVENTED)

$$S_{gauge} = -\frac{1}{4}\int d^8x\sqrt{|g|}\, g^{MP}g^{NQ}\Big(\frac{1}{\hat{g}_s^2}G^\alpha_{MN}G^\alpha_{PQ} + \frac{1}{\hat{g}^2}W^a_{MN}W^a_{PQ} + \frac{1}{\hat{g}'^2}B_{MN}B_{PQ}\Big),$$

$$S_f = \sum_{i=1}^{3}\sum_{\psi}\int d^8x\sqrt{|g|}\Big[\tfrac{1}{2}(\bar{\psi}_i\gamma^M D_M\psi_i - (D_M\bar{\psi}_i)\gamma^M\psi_i) - \mu_{\psi i}\bar{\psi}_i\psi_i\Big],$$

$$D_M = \partial_M + \Omega_M - iG^\alpha_M T^\alpha - iW^a_M\frac{\tau^a}{2} - iYB_M,$$

for three generations $i = 1, 2, 3$ of Q_i ($(u, d)_L$, $(c, s)_L$, $(t, b)_L$), u_{Ri} (u_R, c_R, t_R), d_{Ri} (d_R, s_R, b_R), L_i ($(\nu_e, e)_L$, $(\nu_\mu, \mu)_L$, $(\nu_\tau, \tau)_L$, the neutrinos here the weak partners, not the mass states ν_1, ν_2, ν_3), e_{Ri} (e_R, μ_R, τ_R) and ν_{Ri} (hypercharges $1/6, 2/3, -1/3, -1/2, -1, 0$), each a 16-component spinor of the same Clifford algebra times its gauge indices, with one doublet bulk mass per sector ($\mu_{Li} = \mu_L$, $\mu_{Qi} = \mu_Q$; ASSUMED). With the selection projector

$$P_{sel} = \frac{1 + \Gamma_4\gamma^{x_0}}{2}\,\frac{1 - i\gamma^{x_5}\gamma^{x_6}}{2}, \qquad \Gamma_4 = i\gamma^{x_1}\gamma^{x_2}\gamma^{x_3}\gamma^{x_4},$$

the conditions at the fold and the tip are, for doublets ($\mu_\psi > 0$), $P_{sel}(1 - \gamma^{x_0})\psi = 0$, $(1 - P_{sel})(1 + \gamma^{x_0})\psi = 0$ at the fold and $(1 - \gamma^{x_0})\psi = 0$ at the tip, and for singlets ($\mu_\psi < 0$) the same with $\gamma^{x_0} \to -\gamma^{x_0}$. The weak bosons are $W^\pm = (W^1 \mp iW^2)/\sqrt{2}$, with electric charges $+e$ (W^+) and $-e$ (W^-).

7.4 The tip sigma field and the tip matrices (INVENTED for this geometry; no Higgs particle)

$$S_{tip} = \int_{y=-L} d^7x \sqrt{|g_{tip}|}\Big[-\frac{v_\Sigma^2}{4}\mathrm{tr}\big((D_\mu\Sigma)^\dagger D^\mu\Sigma\big) - \sum_{i,j=1}^{3}\big(\bar{L}_i\,\Sigma\,(\Lambda_\ell)_{ij}\,\ell_{Rj} + \bar{Q}_i\,\Sigma\,(\Lambda_q)_{ij}\,q_{Rj} + \mathrm{h.c.}\big)\Big], \qquad \Sigma \in SU(2),$$

$$D_\mu\Sigma = \partial_\mu\Sigma - iW_\mu^a\frac{\tau^a}{2}\Sigma + iB_\mu\Sigma\frac{\tau^3}{2}, \qquad \Sigma \to U_L\Sigma U_Y^\dagger, \quad U_Y = e^{i\theta\tau^3/2},$$

with the singlets arranged as $SU(2)_R$-type doublets $\ell_{Rj} = (\nu_{Rj}, e_{Rj})$, $q_{Rj} = (u_{Rj}, d_{Rj})$ and $(\Lambda_\ell)_{ij} = \mathrm{diag}((\lambda_\nu)_{ij}, (\lambda_e)_{ij})$, $(\Lambda_q)_{ij} = \mathrm{diag}((\lambda_u)_{ij}, (\lambda_d)_{ij})$ with four complex 3×3 tip coupling matrices λ_ν, λ_e, λ_u, λ_d (the upper component of $\Sigma^\dagger L_i$ couples to ν_R, the lower one to e_R). On the zero modes they become the tip conditions $b_L = \kappa b_R$, $a_R = -\kappa^\dagger a_L$ with the tip matrices (ASSUMED values)

$$\kappa_\nu = U_{PMNS}, \qquad \kappa_e = 1, \qquad \kappa_u = 1, \qquad \kappa_d = V_{CKM}.$$

7.5 The portal interactions (INVENTED)

$$S_{I11} = \int d^8x\sqrt{|g|}\,\frac{c_V}{\Lambda_\star^6}(i\bar{\Psi}\gamma^M\Psi)\sum_{i=1}^{3}\sum_{\psi} q_\psi(i\bar{\psi}_i\gamma_M\psi_i),$$

$$S_{I12} = -\int d^8x\sqrt{|g|}\,\frac{\bar{\Psi}\Psi}{\Lambda_\star^3}\big[c_G G^\alpha_{MN}G^{\alpha MN} + c_W W^a_{MN}W^{aMN} + c_B B_{MN}B^{MN}\big],$$

$$S_{I13} = \int d^8x\sqrt{|g|}\,\frac{c_A}{\Lambda_\star^6}\sum_{M\neq x_0}(i\bar{\Psi}\gamma^M P_b\Psi)\,J_M^B, \qquad P_b = \frac{1-\gamma^{x_0}}{2}, \qquad J_M^B = \frac{1}{3}\sum_{i=1}^{3}\sum_{\psi\in\{Q_i,\,u_{Ri},\,d_{Ri}\}} i\bar{\psi}\gamma_M\psi,$$

with the $B-L$ charges $q_Q = q_{u_R} = q_{d_R} = \frac{1}{3}$, $q_L = q_{e_R} = q_{\nu_R} = -1$ in I11 (version 3; version 2 left the charges general and set $c_V = 0$). I13 is new in version 3: the x_0-chiral current $i\bar{\Psi}\gamma^M P_b\Psi$ was chosen because it is Hermitian for the seven directions $M \neq x_0$ (for $M = x_0$, $(iC\gamma^{x_0}P_b)^\dagger = iC\gamma^{x_0}P_a$, so the sum leaves that component out; item 24), a four-dimensional vector, diagonal in the halves on the zero modes and zero on the fold bands (so it cannot produce dark radiation), and the baryon-number current because the $B-L$ current would thermalise the right-handed neutrinos (item 23). On the zero modes the portals become the four-dimensional couplings $G_V q_f \tilde{I}_{of}$ and $G_A q_f^B \tilde{I}_{of}^{(b)}$ with $G_V = c_V H/(2f^4 v_t \Lambda_\star^6)$, $G_A = c_A H/(2f^4 v_t \Lambda_\star^6)$ and the overlaps $\tilde{I}_{of} = \int_{-L}^0 (a_o^2 + b_o^2)\varrho_f e^{-3Hy}dy$, $\tilde{I}_{of}^{(b)} = \int_{-L}^0 b_o^2\varrho_f e^{-3Hy}dy$ (DERIVED). Removed in version 2 (DERIVED reasons in item 21): the chiral tip Z portal of version 1,

$$S_{I9} = \int_{y=-L} d^7x\sqrt{|g_{tip}|}\big[c_Z + c_Z'\varsigma\big](i\bar{\Psi}\gamma^\mu P_+\Psi)\,\frac{iv_\Sigma^2}{2}\mathrm{tr}\big(\Sigma^\dagger D_\mu\Sigma\,\tau^3\big), \qquad P_+ = \frac{1+\Gamma_4}{2}, \quad \varsigma = \pm1,$$

now $c_Z = c_Z' = 0$. Operators considered and rejected: a dipole portal $\bar{\Psi}S^{MN}\Psi B_{MN}$ (it would thermalise the sterile fold bands through the photon: too much dark radiation) and any operator coupling the neutral Ψ to a single W^+ or W^- (none is gauge invariant).

7.6 Parameter choices (ASSUMED)

$\eta = 3$ and $m = H$ (from the record); the universal tip mixing $\kappa_f = 1$ (unitary tip matrices); the doublet localisations fixed by the heaviest member of each sector, $c_R(\tau) = -c_L$ and $c_R(t) = -c_Q$; the lightest neutrino mass 1 meV and the normal ordering; the Kohn-Sham benchmark self-coupling $\lambda_B = 7\times10^{-5}H^{-6}$ (mean fields below 5 % of m in the densest benchmark state; it is only the value with which the Kohn-Sham machinery is tested, see below); the extra-time coordinate volume v_t (it enters only $V_\perp$ and the conversion of c_V, c_A and λ into G_V, G_A and G_λ). Version 3: the I11 strength $G_V = 4.38\times10^{-17}$ GeV^{-2}, fixed by the freeze-in benchmark $\Delta N_{eff} = 0.001$ from the highest temperature $T_{max} = 10$ TeV of the hot Universe (both ASSUMED; version 2: $c_V = 0$); the I12 strength $G = \pi/(2M_{DM}^3)$ with universal coefficients (ASSUMED, chosen to respect the $^3P_0$ unitarity bound up to $\sqrt{s} = 2.4M_{DM}$); the I13 strength is DERIVED from $\Omega_c h^2 = 0.120$; the self-interaction strength $G_\lambda = \lambda H\tilde{I}_\lambda/(2f^4 v_t) \leq 2.3\times10^{-15}$ GeV^{-2} (ASSUMED bound from $\Delta N_{eff} \leq 0.001$ of the channel $\chi\bar{\chi} \to$ fold pairs, item 24; $\lambda \leq 1.4\times10^{-10}H^{-6}$ for $v_t = H^{-3}$).

8 The computations and their verification (DERIVED)

All computations are Python scripts in the folder `universe_model_2/code`; each writes its numbers to `universe_model_2/outputs` and a report of named checks (name, verdict, detail) to `universe_model_2/reports`. In total 235 checks in version 3 (233 before the fixes of the independent review, item 24; 205 in version 2), none failed.

script	what it computes	method	checks
`um2_algebra.py`	the author's gamma matrices rebuilt from the notebook formulas; C, Γ, B; Pin(4,4) irreducibility; the two inequivalent Spin(4,4) halves; the 2×2 block basis and the block forms of every matrix the reductions use	sympy, exact	31/31
`um2_geometry.py`	Riemann tensor of $\bar{g}$ in two charts; Lovelock tensors by four routes (pairing formula, Ricci, explicit Gauss-Bonnet tensor, minisuperspace variation with a lapse); Bianchi identities; the record's equations; the static block; the canonical spin connection	sympy, exact	25/25
`um2_background.py`	the class-form background equations; the static block; the stabiliser and its family; the radion; the linear response of the block to matter sources; the dimensional reduction	sympy, exact; eigenvalues numerically	22/22
`um2_fields.py`	the exact x_0 reduction of the eight-dimensional Dirac equation (all blocks, general a_4, b_4, $M(y)$, $v(y)$); boundary conditions; zero modes; tip mixing for one and for three generations (self-adjointness with an arbitrary complex 3×3 matrix); the gauge x_0 operator; gauge invariance of the three-generation tip terms	sympy and numpy	16/16

script	what it computes	method	checks
`um2_spectra.py`	the x_0 spectra of gauge bosons and graviton; the $W^{\pm}$, Z fit to ρ_0 with the angle of the couplings fixed by the measured masses (both masses reproduced); the twelve fermion masses as exact x_0 eigenvalues	mpmath root finding; quadratic finite elements with Richardson; ODE shooting	14/14
`um2_flavour.py`	the PMNS and CKM matrices; the exact three-generation tip determinant and its null vectors; a general complex tip matrix; flavour-changing couplings of the Kaluza-Klein gauge modes; $\mu \to e\gamma$; lepton universality	mpmath (50 digits), numpy	7/7
`um2_neutrino.py`	vacuum and matter oscillations, the MSW resonances, long-baseline appearance, the Sun, the onset of oscillations in the early Universe, neutrino decoupling and N_{eff}, the CNB, supernova resonances, the fast flavour instability, the heating behind the shock; (version 3) the Majorana alternative	exact matrix exponentials and layer products; LSODA; Gauss-Legendre panels; vectorised Newton iterations	16/16
`um2_ks_jobs.py`	Kohn-Sham: validation against the record (2/2), free spectrum (3/3), interacting ground states $N = 8$ (3/3) and $N = 592$ (4/4), dark-matter quanta (3/3), portal responses with the three-generation sources (5/5), block feedback with the full adiabatic response (2/2)	Galerkin in the free basis, safeguarded Anderson self-consistency, Richardson in the basis size	22/22
`um2_electroweak.py`	charges (W^+: $+e$, W^-: $-e$), anomalies of three generations, couplings, P_+, $\Gamma_4 = -B\gamma^{x_0}$, chiral tip charges of Ψ_A, Ψ_B, N_ν, the removal of the tip Z portal, the Z widths of three generations and their scheme dependence	sympy and numerics	19/19

script	what it computes	method	checks
`um2_primordial.py`	the nonlinear primordial evolution, the basin, the separatrix, the radion oscillation	DOP853 integration, bisection	5/5
`um2_cosmology.py`	TT and EE spectra with three massive neutrinos, BAO χ^2, N_{eff}, helium, the attribution of the differences, the variants with a thermalised fold band; (version 3) the model with the freeze-in fold bands, σ_8, $f\sigma_8$, the potential spectrum	CAMB 2.0.5	8/8
`um2_analysis.py`	the x_0 decompositions and the uncertainty relation on the cell (with its tip term); the gap theorem; sum rules; the twelve fermions against the uncertainty estimate; the conservation of the halves Ψ_A, Ψ_B; relic abundance of two species; direct detection; the condensate; the vacuum energy; the sensitivities to the block size (with the neutrino splittings); the bound on the block rate today; the radion decay	mpmath, numpy, sympy	23/23
`um2_thermal.py` (version 3)	the degrees of freedom with the x_0 masses; the relativistic thermal average; the Boltzmann solver and its validation; the relic of I12 alone; partial-wave unitarity (the ${}^3P_0$ limit, the capped operator); the unitarity-safe I12 benchmark	Gauss-Legendre ideal-gas integrals; adaptive quadrature with scaled Bessel functions; implicit Radau; Brent root finding	7/7

script	what it computes	method	checks
`um2_coupled.py` (version 3)	the portal overlaps with every zero mode and their Kohn-Sham checks; the screened response; the fold band under the portals; the freeze-in of the fold bands; direct detection; I13 (the algebra of its bilinear with the author's matrices, fold-band blindness, relic with unitarity cap, detection, annihilation today); the $B-L$ alternative and the thermalised ν_R; the self-interaction channel and its bound; the mean fields in ten environments (with the order-α electroweak condensates); the full adiabatic block response to every species and the proton sensitivity; the W eigenvalue with a kinetic function; the domain of validity	exact sympy matrices; self-consistent Kohn-Sham solutions with Richardson in the amplitude; Boltzmann equations; quadratic finite elements	16/16
`um2_series.py` (version 3)	the magnitudes of the terms of the metric series: potentials, Kaluza-Klein gravitons (levels, norms, Yukawa strengths, orthogonality), the radion, order 2, tensor bound	analytic eigenfunctions, finite elements, quadrature	4/4

8.1 The principal numbers and where they come from

quantity	value	computed by
static block k_*, f_*, $G(k_*)$, $\kappa\rho_b^*$	$12+\frac{4}{3}\sqrt{51}=21.5219$, 0.215556, $\frac{20}{3}+\sqrt{51}=13.8081$, $(26+4\sqrt{51})H^2$	`um2_background.py`
radion	$\omega_r^2/H^2=31552/2249-2416\sqrt{51}/6747=11.4721$, $m_r=3.387H=48.42$ TeV	`um2_background.py`, `um2_spectra.py`
unbroken x_0 spectrum ($\eta=3$)	$\Lambda=0$; $\Lambda_1=30.1196$; odd 27.0890; gap 20.25	`um2_spectra.py`
$W^{\pm}$, Z	$\beta_W=0.011912$, $\beta_Z=0.015343$ ($\cos^2\theta_W=0.776368$), $\Lambda_W=1.32149\times10^{-5}$, $\Lambda_Z=1.70150\times10^{-5}$, $\rho-1=0.00038$, model $m_Z=91.1879$ GeV; Dirichlet bound $\Lambda_D=0.010016$	`um2_spectra.py`

quantity	value	computed by
scales	$m_{KK} = 121.32$ TeV, $H = 14.296$ TeV, proper half cell $fL = 8.926 \times 10^{-21}$ m, x_0 period 7.227×10^{-21} m, gap 99.48 TeV	`um2_spectra.py`
fermion localisations ($\kappa_f = 1$)	doublets $c_L = 5.9529$, $c_Q = 3.3823$; singlets c_R: ν_1 -35.0452, ν_2 -32.8481, ν_3 -31.0702, e -14.5539, μ -8.9809, τ -5.9529, u -15.4300, c -8.7683, t -3.3823, d -14.6258, s -11.5158, b -7.5005	`um2_spectra.py`
three generations	tip determinant roots equal the one-generation roots to 3.0×10^{-42}, null vectors the columns of κ to 1.1×10^{-16}; general complex tip matrix: singular values to 3.5×10^{-5}	`um2_flavour.py`
mixing	PMNS ($\theta_{12} = 33.71°$, $\theta_{13} = 8.56°$, $\theta_{23} = 43.28°$, $\delta = 212°$) unitary to 1.1×10^{-16}, $J = -0.01777$; CKM $J = 3.117 \times 10^{-5}$; Kaluza-Klein gauge overlaps 0.14434 (leptons), -0.30066 (quarks), off-diagonal 1.0×10^{-17}; $\mathrm{BR}(\mu \to e\gamma) = 2.8 \times 10^{-55}$	`um2_flavour.py`
neutrino masses	1.000, 8.712, 50.140 meV; $\sum m_\nu = 0.0599$ eV	`um2_spectra.py`, `um2_neutrino.py`
flavour conversion	1-3 MSW resonance 11.135 GeV (exact 11.133 GeV) at 2.8 g/cm^3; solar 1-2 resonance 1.93 MeV; $P(\nu_\mu \to \nu_e)$: T2K 0.0517, NOvA 0.0481, DUNE 0.0701 (antineutrinos 0.0313, 0.0236, 0.0195); solar P_{ee} from 0.5408 (pp) to 0.3043 (^{8}B, 15 MeV)	`um2_neutrino.py`
early Universe	oscillation onset $T = 4.99$ MeV (atmospheric), 2.39 MeV (solar); $N_{eff} = 3.0476$ (separate temperatures), 3.0485 (common); adopted 3.0432	`um2_neutrino.py`
CNB	$T_\nu = 1.9454$ K, 112.0 cm^{-3} per mass state, $\langle v \rangle = 0.435$, 0.0604, 0.0105; $\Omega_\nu h^2 = 6.382 \times 10^{-4}$	`um2_neutrino.py`
supernovae	H resonance (20 MeV) at 1573 g/cm^3, $\gamma = 178$; $P_{ee} = 0.02215$, $\bar{P}_{ee} = 0.677$; fast instability growth 0.0877μ and 0.0979μ, e-folding 5.9×10^{-10} to 5.9×10^{-11} s; heating factors 0.884, 0.760	`um2_neutrino.py`
dirac16complex at f_*	$c_\psi = 0.646667$; $\varepsilon_{DM} = 3.102204\,m$ ($p = 2.93661\,m$); even bulk $4.959981\,m$; x_0-momentum fraction 0.896	`um2_ks_jobs.py spectrum`

quantity	value	computed by
dark matter	$M_{DM} = 44.35$ TeV (both halves); $w = 0$; $p_0/\rho = 0.772034343$ $= -d\ln\varepsilon_{DM}/d\ln f$	`um2_ks_jobs.py darkmatter`
portal responses $d\varepsilon_{odd}/de$	I11 electron 0.824779, up quark 4.328803, neutrino 1.252406; I12 gluon condensate 0.707196	`um2_ks_jobs.py portals`
tip Z portal (removed)	$N_\nu = 3 + 2r^2$, $r^2 < 0.0056$; at the bound decoupling at 14.7 MeV and $\Delta N_{eff} = 2.02$; oriented part $\Delta N_{eff} = 0.0958$ to 0.1257	`um2_electroweak.py`
Z widths	invisible 497.7 MeV (measured 499.1), total 2422.3 MeV tree level, 2485.2 MeV with $1 + \alpha_s/\pi$ (measured 2495.5); $R_\ell = 20.83$ (measured 20.77)	`um2_electroweak.py`
primordial era	separatrix $k_c = 2.6983, 2.5991, 2.5301$; deflation factor up to 2.9166; radion $3.3868H$ measured	`um2_primordial.py`
cosmology	UM-2 (version 3, $N_{eff} = 3.0442$): $H_0 = 67.38$ km/s/Mpc, $\Omega_m = 0.3150$, $r_{drag} = 147.104$ Mpc, BAO $\chi^2 = 31.49$ (13 points; ΛCDM 32.55), TT and EE within 2.5×10^{-4} and 2.7×10^{-4} of ΛCDM; without the freeze-in fold bands (the neutrino content of version 2, $N_{eff} = 3.0432$): 67.37, 0.3151, 147.109 Mpc, 31.72, 2.8×10^{-4}	`um2_cosmology.py`
relic	two species, $x_f = 30.38$, $g_* = 105.75$, required $\langle\sigma v\rangle = 2.01 \times 10^{-25}$ cm^3/s per species, $(8G_g^2 + 3G_W^2 + G_B^2)^{1/2} =$ $(15.5\ \text{TeV})^{-3}$; the $^3P_0$ unitarity bound $\sqrt{\pi x_f}/M^2 = 5.79 \times 10^{-26}$ cm^3/s is exceeded 3.47 times ($v_c = 0.263$, 79 % of the pairs above it): relic OPEN in version 2 (I12 alone), resolved in version 3 by I13	`um2_analysis.py`
direct detection	$\sigma_{SI} = 6.3 \times 10^{-53}$ cm^2 (gluon-only) against 2.4×10^{-45} cm^2	`um2_analysis.py`
dark energy	condensate tuned to $w_0 = -1$: $w_a = -6$, energy zero at $z = 0.260$; vacuum-energy ratio 5.5×10^{64}; $(H_0/H)^2 = 1.0 \times 10^{-92}$	`um2_analysis.py`

quantity	value	computed by
uncertainty relation	m/m_H: first KK 7.49, first odd 6.47, dark matter 2.29, $W^{\pm}$ 1.14×10^{-3}, Z 1.29×10^{-3}, top quark 3.2×10^{-3}, electron 5.5×10^{-9}, ν_1 1.1×10^{-17}; $\langle p^2 \rangle \Delta\xi^2$: photon zero mode 0.497, Ψ fold zero mode 0.097 ($\varrho(\text{tip}) = 0.489$); $\sqrt{\langle p^2 \rangle}\Delta\xi \geq (1 - \varrho(\text{tip}))/2$ for every tabulated state	`um2_analysis.py`
sensitivities $d \ln X / d \ln f$	KK -1, m_W -0.501, fermions from -6.8 (t) to -41.0 (ν_1), m_e -20.5, M_{DM} -0.772, α and the gauge couplings -4, G_N $+2.05$; splittings Δm^2_{21} -77.5, Δm^2_{31} -74.0	`um2_analysis.py`
block rate today (ESTIMATE)	$\lvert h \rvert < 3.6 \times 10^{-8} H_0$ (from $\lvert \dot{\alpha}/\alpha \rvert < 10^{-17}$/yr), $< 7.1 \times 10^{-4} H_0$ (from $\lvert \dot{G}/G \rvert < 10^{-13}$/yr)	`um2_analysis.py`
radion decay (ESTIMATE)	$\Gamma = C_r \, m_r^3 / (4\pi \bar{M}_{Pl}^2)$, $T_{RH} = (90/(\pi^2 g_*))^{1/4} \sqrt{\Gamma \bar{M}_{Pl}}$ ($g_* = 10.75$): $\tau = 0.43\,\text{s}/C_r$, $T_{RH} = 1.85\sqrt{C_r}$ MeV (< 4.7 MeV unless $C_r > 6.5$)	`um2_analysis.py`
first-order block response	$s_{h3} = 0.028666$, $s_{hp} = 0.037320$, $q_h = -0.009555$; adiabatic $\delta k = (-0.107556\,\kappa\delta p_3 - 0.140028\,\kappa\delta\bar{p} + 0.035852\,\kappa\delta\rho)/H^2$ (radiation drops out: $0.107556 = 3 \times 0.035852$); for the Kohn-Sham states $\delta k = 2.27 \times 10^{-6}\kappa$ ($N = 8$, $+\lambda_B$), 0.0125κ ($N = 592$, $+\lambda_B$), 0 (free gas)	`um2_background.py`, `um2_ks_jobs.py feedback`
thermal history (version 3)	$g_\rho = 105.75$ (5 TeV), 105.73 (1.42 TeV), 104.95 (200 GeV), 86.22 (10 GeV), 10.74 (3 MeV); today $g_s = 3.9091$, $g_\rho = 3.3626$, $s_0 = 2891.3$ cm^{-3}; thermal average $(1 + c_1/x)$, $c_1 \approx 3$	`um2_thermal.py`
I12 relic and unitarity (version 3)	(15.855 TeV)$^{-3}$, $x_f = 31.70$, $\langle \sigma v \rangle(x_f) = 1.891 \times 10^{-25}$ cm^3/s; crossing of the $^3P_0$ bound at $\sqrt{s} = 89510$ GeV, $v = 2\beta = 0.269$ (0.274 from the non-relativistic formula), 96 % of the rate and 75 % of the pairs above it; $^3P_0$ limit: 0.0582 per species ($x_f = 30.77$, $\langle \sigma v \rangle = 5.33 \times 10^{-26}$ against $\sqrt{\pi x}/M^2 = 5.84 \times 10^{-26}$ cm^3/s); benchmark fraction above its crossing 5.2×10^{-4}; capped: 0.1544, 0.1081, 0.0836, 0.0649, 0.0585; benchmark (38.2 TeV)$^{-3}$ alone 8.0; $c/\Lambda_\star^3 = 1.106 \times 10^{-14}$ GeV^{-3}	`um2_thermal.py`

quantity	value	computed by
I13 relic (version 3)	$\sum_f N_c(q_f\tilde{I}_f^{(b)})^2 = 16.351$; $G_A = 1.278\times10^{-9}$ GeV^{-2} (capped; 1.269×10^{-9} without the cap); $x_f = 31.23$; $\langle\sigma v\rangle(x_f)$: I13 9.50×10^{-26}, I12 9.9×10^{-28} cm^3/s; bound reached above $v = 0.51$ (22 % of the rate at the bound; the uncapped cross-section crosses it above $v = 0.52$, 25 % of the rate); $B-L$ alternative $G_A^{B-L} = 1.072\times10^{-9}$ GeV^{-2}, ν_R thermalised, kinetic decoupling at $M/T = 24.4$, $\Delta N_{eff} = 0.144$; with J^B the ν_R production through I11 is 2.0×10^{-17} of the I13 rate	`um2_coupled.py`
self-interaction channel (version 3, after the review)	$\tilde{I}_\lambda = 14.38$; $\Delta N_{eff} = 1.83\times10^{26}\,G_\lambda^2$ GeV^4; bound $G_\lambda \le 2.3\times10^{-15}$ GeV^{-2} ($1.8\times10^{-6}G_A$); λ_B with $v_t = H^{-3}$: $G_\lambda = 1.14\times10^{-9}$ GeV^{-2}, fold bands thermalised to 2.6 TeV, $\Delta N_{eff} = 0.19$	`um2_coupled.py`
freeze-in and detection (version 3)	$\sum_f N_c(q_f\tilde{I}_f)^2 = 126.85$; $\Delta N_{eff} = 5.21\times10^{29}G_V^2$ GeV^4; $G_V = 4.38\times10^{-17}$ GeV^{-2}; $\Gamma/H = 0.012$ at 10 TeV; xenon $\sigma_n = 4.01\times10^{-46}$ cm^2 (0.16 of LZ); σv today 9.76×10^{-26} cm^3/s	`um2_coupled.py`
coupled solution (version 3)	overlaps $\tilde{I}_Q = 8.2722$, $\tilde{I}_Q^{(b)} = 2.6135$ (dark matter); Kohn-Sham checks 3.1×10^{-10}, 2.3×10^{-10}; the I13 algebra exact; $\delta\varepsilon/M \le 1.5\times10^{-6}$ (freeze-out at $T = 1.42$ TeV; thermal condensates -8.1×10^{12}, -5.0×10^{11}, -2.6×10^{10} GeV^4; Feynman-Hellmann against the trace anomaly 10^{-16}); $\delta\ln f \le 1.1\times10^{-34}$ (the hot plasma at the freeze-out; 7×10^{-39} in the electroweak era, 9×10^{-49} from nucleosynthesis on); loop gain $\le 1.1\times10^{-33}$; $d\ln m_p/d\ln f = -46.1$; amplitude $e = 1.7\times10^{-16}$ at the freeze-out; $d\ln\Lambda_W/de = -0.1126$; validity up to $T = 27.4$ TeV	`um2_coupled.py`

quantity	value	computed by
metric series and σ_8 (version 3)	$\Delta_\Phi = 2.42 \times 10^{-5}$ ($k = 10^{-4}$ Mpc^{-1}) to 5.03×10^{-8} (1 Mpc^{-1}); order 2: 5.8×10^{-10}; $g(0) = 0.7878$; graviton levels 30.1196, 59.7284, 109.0764; $\alpha_n = 0.0728$, 0.1469, 0.1809 (two masses at the fold); range 1.63×10^{-21} m; radion $\leq 1.1 \times 10^{-34}$; $\sigma_8 = 0.8118$ (UM-2, version 3), 0.8119 (without the freeze-in fold bands), 0.8109 (ΛCDM)	`um2_series.py`, `um2_cosmology.py`
Majorana alternative (version 3)	$\kappa_M = 5.63 \times 10^{-13}$, 4.90×10^{-12}, 2.82×10^{-11}; $m_{\beta\beta}$ from 0.84 to 4.41 meV	`um2_neutrino.py`

8.2 Methods, with the equations they use

- **Exact checks.** Symbolic identities are verified with sympy by simplifying the difference to zero; matrix identities entry by entry.
- **Curvature and Lovelock tensors.** For the order-0 metric the Riemann tensor is pair-diagonal, $R^{ab}{}_{cd} = 0$ unless $\{a, b\} = \{c, d\}$, with the sectional curvatures $K_{ab} = R^{ab}{}_{ab}$ ($i, j \in \{1, 2, 3\}$; $B, B' \in \{0, 5, 6, 7\}$; $A = \dot{a}_4$, $h = \dot{f}/f$, $k = 1/f^2$)

$$K_{ij} = A^2, \quad K_{i4} = A^2 + \dot{A}, \quad K_{iB} = Ah, \quad K_{4B} = h^2 + \dot{h}, \quad K_{BB'} = h^2 - H^2 k,$$

  and the Lovelock densities and tensors follow from the pairing formula

$$\mathcal{L}_{(k)} = 2^k k! \sum_{\text{k disjoint pairs}} \prod_{\text{pairs}} K_{ab}, \qquad E_{(k)}{}^h{}_h = -2^{k-1} k! \sum_{\text{k disjoint pairs avoiding } h} \prod_{\text{pairs}} K_{ab},$$

  checked against the Einstein tensor from the Ricci tensor ($k = 1$), the explicit Gauss-Bonnet tensor ($k = 2$), the Bianchi identity $\nabla_M E_{(k)}{}^M{}_N = 0$ and the Euler-Lagrange equations of the action with a lapse ($k = 1, 2, 3$).
- **Boson eigenvalues.** With $\xi = y/L$, $u = e^{-3\eta\xi/2}\tilde{u}$, $q^2 = \Lambda - 9\eta^2/4$, the operator $-e^{-3\eta\xi}(e^{3\eta\xi}u')' = \Lambda u$ with $u'(0) = 0$ (even) or $u(0) = 0$ (odd) and $u'(-1) = \beta u(-1)$ has the eigenvalue conditions $\left[q^2 + \frac{3\eta}{2}(\beta + \frac{3\eta}{2})\right] \sin q - \beta q \cos q = 0$ (even) and $q \cos q + (\beta + \frac{3\eta}{2}) \sin q = 0$ (odd), solved with mpmath at 30 digits. Cross-check: quadratic finite elements for $\int e^{3\eta\xi} u'^2 d\xi + \beta e^{-3\eta} u(-1)^2 = \Lambda \int e^{3\eta\xi} u^2 d\xi$ on 100, 200, 400 elements, the eigenvalue taken from the Rayleigh quotient of the gradient form and extrapolated by Richardson.
- **Richardson extrapolation.** For a quantity with error $C\, n^{-p}$: $x \approx x_{2n} + (x_{2n} - x_n)/(2^p - 1)$; the observed order is $\log_2[(x_n - x_{2n})/(x_{2n} - x_{4n})]$.
- **Fermion eigenvalues.** The exact condition $D(e) = e^2 s_L s_R - \kappa_f^2 (C_L - c_L s_L)(C_R + c_R s_R)$, $s = \sinh w/w$, $C = \cosh w$, $w = \sqrt{c^2 - e^2}$, solved with mpmath; the first-order solution $e = \kappa_f \zeta(c_L)\zeta(-c_R)$, $\zeta(c)^2 = c\, e^{-c}/\sinh c$ ($c_L > 0$, $c_R < 0$); cross-check by direct integration of the coupled (L, R) Dirac system (DOP853, relative tolerance 10^{-13}). With the doublet value fixed by the heaviest member of a sector ($c_R = -c_L$ for τ and t), each lighter c_R is found by root finding on the mass.
- **Three generations.** With the boundary values $(a_L, b_L) = (C_L - c_L s_L, -s_L e)$, $(a_R, b_R) = (s_R e, C_R + c_R s_R)$ at the tip, the conditions $b_L = \kappa b_R$, $a_R = -\kappa^\dagger a_L$ give the 3×3 problem $\left[\mathrm{diag}(b_{L\alpha}) + \kappa\, \mathrm{diag}(b_{Ri}/a_{Ri})\, \kappa^\dagger\, \mathrm{diag}(a_{L\alpha})\right]\mathcal{A} = 0$, whose determinant is evaluated with mpmath at 50 digits and whose roots are found by the Illinois (modified regula falsi) method in a ± 2 % bracket around the one-generation roots; the null vectors (right singular vectors of the smallest singular value) are compared with the columns of κ. For a non-unitary random tip matrix the masses are compared with the singular values of $\zeta(c_L)\, \kappa\, \mathrm{diag}(\zeta(-c_{Ri}))$.

- **Flavour evolution.** With $\mathcal{H} = U\,\mathrm{diag}(m_1^2, m_2^2, m_3^2)\,U^\dagger/(2E) + \mathrm{diag}(V_{CC}, 0, 0)$, $S = e^{-i\mathcal{H}L}$ by diagonalisation of the Hermitian $\mathcal{H}$ for constant density and $P(\nu_\alpha \to \nu_\beta) = |S_{\beta\alpha}|^2$; the ordered product of layer operators (20000 and 40000 layers) through the Sun with $n_e(r) = N_A \min(245\, e^{-10.54\, r/R_\odot}, 100)$ cm^{-3}, followed by the incoherent sum $P_{ee} = \sum_i |U_{ei}|^2 P(\nu_i)$, compared with the adiabatic formula

$$P_{ee} = c_{13}^4\Big(\frac{1}{2} + \frac{1}{2}\cos 2\theta_{12}\cos 2\theta_{12}^m(r_0)\Big) + s_{13}^4,$$

$$\cos 2\theta_{12}^m = \frac{\cos 2\theta_{12} - A}{\sqrt{(\cos 2\theta_{12} - A)^2 + \sin^2 2\theta_{12}}}, \qquad A = \frac{2EV_{CC}(r_0)c_{13}^2}{\Delta m_{21}^2}$$

(r_0 the production radius); the 1-3 resonance condition $2EV_{CC} = \Delta m_{ee}^2 \cos 2\theta_{13}$, $\Delta m_{ee}^2 = c_{12}^2\Delta m_{31}^2 + s_{12}^2\Delta m_{32}^2$, against the exact three-flavour resonance $|U_{e3}^m|^2 = 1/2$ found by Brent's method; the onset of oscillations in the early Universe where the thermal potential of the $e^\pm$ plasma,

$$V_e - V_\mu = -\frac{8\sqrt{2}\,G_F p}{3m_W^2}\,\frac{7\pi^2}{60}T^4,$$

falls below $\Delta m^2\cos 2\theta/(2p)$ at the mean momentum $p = 3.15T$.

- **Supernovae.** In the envelope $\rho(r) = 10^4\,(10^9\,\mathrm{cm}/r)^3$ g/cm^3, $Y_e = 0.5$, a two-flavour resonance (Δm_{31}^2, θ_{13} for H; Δm_{21}^2, θ_{12} for L) lies at $n_e = \Delta m^2\cos 2\theta/(2\sqrt{2}\,G_F E)$, where $d\ln n_e/dr = -3/r$; its adiabaticity and jump probability are

$$\gamma = \frac{\Delta m^2 \sin^2 2\theta}{2E\cos 2\theta\,|d\ln n_e/dr|}, \qquad P_{LZ} = e^{-\pi\gamma/2};$$

with both resonances adiabatic (normal ordering) a ν_e leaves the star as ν_3, $P(\nu_e \to \nu_e) = |U_{e3}|^2$, and an $\bar\nu_e$ as $\bar\nu_1$, $\bar P(\bar\nu_e \to \bar\nu_e) = |U_{e1}|^2$; the observed mean energies are $\langle E\rangle_{\nu_e} = [PN_e\langle E\rangle_e + (1-P)N_x\langle E\rangle_x]/[PN_e + (1-P)N_x]$ (and the same with $\bar P$, $N_{\bar e}$, $\langle E\rangle_{\bar e}$ for $\bar\nu_e$) with $N \propto L/\langle E\rangle$; the schematic charged-current heating $q \propto N_e\langle E^2\rangle_e + N_{\bar e}\langle E^2\rangle_{\bar e}$, $\langle E^2\rangle = \frac{5}{4}\langle E\rangle^2$, becomes $q' = \frac{1}{3}(N_e\langle E^2\rangle_e + 2N_x\langle E^2\rangle_x) + \frac{1}{3}(N_{\bar e}\langle E^2\rangle_{\bar e} + 2N_x\langle E^2\rangle_x)$ under three-flavour equipartition and $q'' = (1 - \Delta/N_e)N_e\langle E^2\rangle_e + (1 - \Delta/N_{\bar e})N_{\bar e}\langle E^2\rangle_{\bar e}$, $\Delta = \frac{2}{3}(N_{\bar e} - N_x)$, under pair conversion.

- **Neutrino decoupling.** The temperature equations of Escudero (MeV units; $H = \sqrt{8\pi\rho/3}/M_{Pl}$ the Hubble rate of the total density, $M_{Pl} = 1.22091\times 10^{19}$ GeV)

$$\frac{dT_\gamma}{dx_4} = -\frac{4H\rho_\gamma + 3H(\rho_e + p_e) + 3HT_\gamma\frac{dP_{int}}{dT_\gamma} + \frac{\delta\rho_{\nu_e}}{\delta x_4} + 2\frac{\delta\rho_{\nu_\mu}}{\delta x_4}}{\frac{d\rho_\gamma}{dT_\gamma} + \frac{d\rho_e}{dT_\gamma} + T_\gamma\frac{d^2P_{int}}{dT_\gamma^2}}, \qquad \frac{dT_{\nu_\alpha}}{dx_4} = -HT_{\nu_\alpha} + \frac{\delta\rho_{\nu_\alpha}/\delta x_4}{d\rho_{\nu_\alpha}/dT_{\nu_\alpha}},$$

$$\frac{\delta\rho_{\nu_e}}{\delta x_4} = \frac{G_F^2}{\pi^5}\Big[4(g_{eL}^2 + g_{eR}^2)F(T_\gamma, T_{\nu_e}) + 2F(T_{\nu_\mu}, T_{\nu_e})\Big], \qquad \frac{\delta\rho_{\nu_\mu}}{\delta x_4} = \frac{G_F^2}{\pi^5}\Big[4(g_{\mu L}^2 + g_{\mu R}^2)F(T_\gamma, T_{\nu_\mu}) - F(T_{\nu_\mu}, T_{\nu_e})\Big],$$

$$F(T_1, T_2) = 32f_a(T_1^9 - T_2^9) + 56f_sT_1^4T_2^4(T_1 - T_2), \qquad f_a = 0.884, \quad f_s = 0.829,$$

with $\rho_\gamma = \pi^2T_\gamma^4/15$, $\rho_{\nu_\alpha} = 7\pi^2T_{\nu_\alpha}^4/120$ per flavour (ν_τ as ν_μ), the $e^\pm$ density and pressure with the electron mass, the effective couplings $g_{eL} = 0.727$, $g_{eR} = 0.233$, $g_{\mu L} = -0.273$, $g_{\mu R} = 0.233$ (or the UM-2 tree-level couplings $\frac{1}{2} + s^2$, s^2, $-\frac{1}{2} + s^2$, s^2, $s^2 = 0.22363$), and the QED interaction pressure $P_{int} = P_2 + P_3$,

$$P_2 = -\frac{e^2T^2}{12\pi^2}J_1 - \frac{e^2}{8\pi^4}J_1^2 + \frac{e^2m_e^2}{16\pi^4}J_2, \qquad P_3 = \frac{e^3T}{12\pi^4}I^{3/2}, \qquad \rho_{int} = -P_{int} + T\frac{dP_{int}}{dT},$$

$$J_1 = \int_0^\infty dp\,\frac{p^2}{E}n_D, \qquad J_2 = \int_0^\infty\int_0^\infty dp\,dq\,\frac{pq}{E_pE_q}\ln\Big|\frac{p+q}{p-q}\Big|n_D(p)n_D(q), \qquad I = \int_0^\infty dp\,\frac{p^2+E^2}{E}n_D,$$

$n_D = 2/(e^{E/T} + 1)$, $E = \sqrt{p^2 + m_e^2}$. The integrals are evaluated with Gauss-Legendre panels, the logarithmic singularity of J_2 removed by $q = p(1 - s^2)$; LSODA from 20 MeV to 5 keV (relative tolerance 10^{-11});

$$N_{eff} = \Big(\frac{11}{4}\Big)^{4/3}\Big[\Big(\frac{T_{\nu_e}}{T_\gamma}\Big)^4 + 2\Big(\frac{T_{\nu_\mu}}{T_\gamma}\Big)^4\Big];$$

for instantaneous decoupling the plasma entropy $a^3(\rho + p)/T$ is conserved and $T_\nu \propto 1/a$.

- **Cosmic neutrino background.** The three mass states with the Fermi-Dirac spectrum $f = 1/(e^{p/T_\nu}+1)$:

$$T_\nu = \Big(\frac{4}{11}\Big)^{1/3} T_{CMB}, \qquad n_i = \frac{3\zeta(3)}{2\pi^2} T_\nu^3 \ (\nu_i + \bar\nu_i), \qquad \rho_i = 2\int \frac{d^3p}{(2\pi)^3}\sqrt{p^2+m_i^2}\, f, \qquad \Omega_\nu h^2 = \frac{\sum_i \rho_i}{\rho_c/h^2},$$

  $\rho_c/h^2 = 1.05368\times 10^4$ eV/cm^3, and the mean velocity $\langle v\rangle = \int p^2 (p/E) f\, dp / \int p^2 f\, dp$, $E = \sqrt{p^2+m_i^2}$.

- **Dark radiation of a thermalised species.** A species of n_W Weyl fermions decoupled when the entropy degrees of freedom were g_s adds

$$\Delta N_{eff} = n_W \Big(\frac{10.75}{g_s}\Big)^{4/3} F_h, \qquad F_h = \Big[\frac{T_\nu}{T_\gamma}\Big(\frac{11}{4}\Big)^{1/3}\Big/(1+h_\nu)\Big]^4 = 1.0095,$$

  with the computed $T_\nu/T_\gamma = 0.71664$ and comoving heating $h_\nu = 0.00165$ of the common-temperature decoupling (F_h corrects for the heating of the neutrinos by e^+e^- annihilation that an earlier-decoupled species does not receive).

- **Fast flavour instability.** With the electron-lepton-number distribution $G(u) = \sqrt{2}G_F[n_{\nu_e} g_{\nu_e}(u) - n_{\bar\nu_e} g_{\bar\nu_e}(u)]$ over the direction cosine u (in units of $\mu = \sqrt{2}G_F n_{\nu_e}$; $\alpha = n_{\bar\nu_e}/n_{\nu_e}$), modes $e^{-i(\Omega x_4 - K x_3)}$ along the symmetry axis x_3, $v = (1,u)$, $\Lambda^\mu = \int du/2\, G\, v^\mu$, $\omega = \Omega - \Lambda^0$, $k = K - \Lambda^3$ and $\eta^{\mu\nu} = \mathrm{diag}(1,-1)$ on (x_4, x_3), the dispersion relation

$$\det \Pi = 0, \qquad \Pi^{\mu\nu} = \eta^{\mu\nu} + \int_{-1}^{1} \frac{du}{2}\, G(u)\, \frac{v^\mu v^\nu}{\omega - ku}$$

  has the axisymmetric branch $(1+J_0)(-1+J_2) - J_1^2 = 0$, $J_n = \int du/2\, G\, u^n/(\omega - ku)$, and the azimuthal branch $-1 + \int du/2\, G(1-u^2)/(2(\omega-ku)) = 0$; it is solved for real k by vectorised Newton iterations with the analytic derivative from a grid of complex starting points; cross-checks against the analytic growth $\frac{1}{2}\sqrt{4I_1^2 - (I_0+I_2)^2}$ ($I_n = \int du/2\, G\, u^n$) of the mode $K = \Lambda^3$ and the eigenvalues of the discretised homogeneous equations $i\, dS_u/dx_4 = (\Lambda^0 - u\Lambda^3)S_u - \int du'/2\, G(u')(1-uu')S_{u'}$ (300 nodes).

- **Kohn-Sham.** The orbitals $h_j\chi_o = \varepsilon_o \chi_o$ (fold $\chi_2(0) = 0$ even or $\chi_1(0) = 0$ odd, tip $\chi_2(-L) = 0$) build the densities

$$n = \sum_o w\, g\, f_o\, n_o, \quad S = \sum_o w\, g\, f_o\, s_o, \qquad n_o = P\chi_o^\dagger \chi_o, \quad s_o = P\chi_o^\dagger j\sigma_2 \chi_o, \quad t_o = P\chi_o^\dagger j\sigma_3\chi_o, \quad P = \frac{e^{-3Hy}}{\mathrm{Vol}_7},$$

  ($w = 1/2$ per fold parity, $g = 4r_3(|\vec n|^2)$ the degeneracy of the momentum shell $\vec k = \Delta k\, \vec n$, r_3 the number of representations as a sum of three squares, f_o the occupations), the energy and the energy-momentum tensor

$$E_{KS} = \sum_o g\, f_o\, \varepsilon_o - 2\,\mathrm{Vol}_7 \int_{-L}^{0} e^{3Hy} e_{int}\, dy, \qquad e_{int} = \frac{15}{32}\lambda S^2 - \frac{1}{32}\lambda n^2,$$

$$\rho = \sum_o w g f_o \varepsilon_o\, n_o - e_{int}, \qquad p_3 = \sum_o w g f_o\, \frac{e^{-a_4|k|}}{3} t_o + e_{int}, \qquad p_t = e_{int},$$

$$p_0 = \sum_o w g f_o \big[(\varepsilon_o - v) n_o - M_{eff} s_o - e^{-a_4|k|} t_o\big] + e_{int},$$

  Orbitals of h_j are expanded in the exact eigenfunctions of the free problem of each sector (60, 120 or 240 functions, quadrature with ten points per function), the mean fields $M_{eff} = m + \frac{15}{16}\lambda S$, $v = -\frac{1}{16}\lambda n$ (plus the portal mean fields $\langle\mathcal{F}\rangle/\Lambda_\star^3$ of I12 and $(c_V/\Lambda_\star^6)\sum_i \sum_\psi q_\psi n_{\psi i}$ of I11 where stated) are iterated to self-consistency (tolerance 10^{-13}) with a safeguarded Anderson mixing and a fixed labelling of the occupied set; checks: the energy identity $E_{KS} = 2\,\mathrm{Vol}_7 \int_{-L}^0 e^{3Hy}\rho\, dy$, the y-conservation law $\big[e^{3Hy} p_0\big]_{-L}^0 = 3H\int_{-L}^0 e^{3Hy} p_t\, dy$, Richardson in the basis size, the Hellmann-Feynman theorem $d\varepsilon/de = \langle\chi|\partial h/\partial e|\chi\rangle$ for every portal mean field, and the reproduction of the record's two reference states to 10^{-13} (energies, HOMO, LUMO) and 10^{-9} (integrated hidden pressure). The portal sources are the zero-mode densities $\varrho_c(y) = (2c/L)e^{2cy/L}/(1-e^{-2c})$: the electron $\frac{1}{2}(\varrho_{c_L} + \varrho_{-c_R(e)})$, the up quark $\frac{1}{2}(\varrho_{c_Q} + \varrho_{-c_R(u)})$, the neutrino ϱ_{c_L}.

- **Primordial era.** The three independent components of the field equations of the order-0 metric with the author's weights and $\Lambda=-10H^2$ (from the pairing formula and the sectional curvatures above; ρ, p_3 and $\bar p=(p_0+3p_t)/4$ the source)

$$
\begin{aligned}
x_4:\quad & -\frac{15A^2h^4}{16H^4}-\frac{5A^3h^3}{4H^4}-\frac{3h^4}{8H^2}-\frac{9Ah^3}{2H^2}-\frac{27A^2h^2}{4H^2}+\frac{9kA^2h^2}{8H^2}\\
&-\frac{3hA^3}{2H^2}+\frac{3hkA^3}{4H^2}-6h^2+\frac{3kh^2}{4}-12Ah+\frac{9Ahk}{2}\\
&-3A^2+\frac{9kA^2}{4}-\frac{3A^2k^2}{16}-10H^2+6kH^2-\frac{3H^2k^2}{8}\\
&=-\kappa\rho\\[2ex]
x_1:\quad & \Big[-\frac{h^4}{8H^4}-\frac{Ah^3}{2H^4}-\frac{3h^2}{2H^2}+\frac{kh^2}{4H^2}-\frac{Ah}{H^2}+\frac{Ahk}{2H^2}\\
&\quad -2+\frac{3k}{2}-\frac{k^2}{8}\Big]\dot A\\
&+\Big[-\frac{Ah^3}{2H^4}-\frac{3A^2h^2}{4H^4}-\frac{3h^2}{2H^2}-\frac{3Ah}{H^2}+\frac{Ahk}{2H^2}-\frac{A^2}{2H^2}\\
&\quad +\frac{kA^2}{4H^2}-4+\frac{3k}{2}\Big]\dot h\\
&-\frac{Ah^5}{2H^4}-\frac{19A^2h^4}{16H^4}-\frac{A^3h^3}{2H^4}-\frac{15h^4}{8H^2}-\frac{6Ah^3}{H^2}+\frac{Akh^3}{2H^2}\\
&-\frac{17A^2h^2}{4H^2}+\frac{7kA^2h^2}{8H^2}-\frac{hA^3}{H^2}+\frac{hkA^3}{2H^2}-10h^2+\frac{9kh^2}{4}\\
&-8Ah+3Ahk-3A^2+\frac{9kA^2}{4}-\frac{3A^2k^2}{16}-10H^2\\
&+6kH^2-\frac{3H^2k^2}{8}\\
&=\kappa p_3\\[2ex]
x_0:\quad & \Big[-\frac{3Ah^3}{8H^4}-\frac{9A^2h^2}{16H^4}-\frac{9h^2}{8H^2}-\frac{9Ah}{4H^2}+\frac{3Ahk}{8H^2}-\frac{3A^2}{8H^2}\\
&\quad +\frac{3kA^2}{16H^2}-3+\frac{9k}{8}\Big]\dot A\\
&+\Big[-\frac{9A^2h^2}{16H^4}-\frac{3hA^3}{8H^4}-\frac{3h^2}{8H^2}-\frac{9Ah}{4H^2}-\frac{9A^2}{8H^2}+\frac{3kA^2}{16H^2}\\
&\quad -3+\frac{3k}{8}\Big]\dot h\\
&-\frac{9A^2h^4}{16H^4}-\frac{17A^3h^3}{16H^4}-\frac{9A^4h^2}{16H^4}-\frac{3h^4}{8H^2}-\frac{27Ah^3}{8H^2}-\frac{45A^2h^2}{8H^2}\\
&+\frac{3kA^2h^2}{16H^2}-\frac{27hA^3}{8H^2}+\frac{9hkA^3}{16H^2}-\frac{3A^4}{8H^2}+\frac{3kA^4}{16H^2}-6h^2\\
&+\frac{3kh^2}{8}-9Ah+\frac{9Ahk}{8}-6A^2+\frac{9kA^2}{4}-10H^2\\
&+3kH^2\\
&=\kappa\bar p
\end{aligned}
$$

  are solved for $\dot A$, $\dot h$ (the 3-space and block components are linear in them), with $\dot k=-2kh$ and the stabiliser $\kappa\rho_b=\kappa\rho_b^*(k/k_*)^4$, $p_3=-\rho_b$, $\bar p=\rho_b$ (plus radiation where stated); the x_4 constraint fixes the initial A and is monitored (residual below 6×10^{-10}); the separatrix is found by bisection in the initial k.

- **Cosmology.** The linear perturbations obey the Einstein-Boltzmann equations (Newtonian gauge, conformal time τ with $dx_4=a\,d\tau$, $\mathcal H=a'/a$, $ds^2=a^2[-(1+2\psi)d\tau^2+(1-2\phi)d\vec x^{\,2}]$, a Fourier mode k; Ma and Bertschinger)

$$k^2\phi+3\mathcal H(\phi'+\mathcal H\psi)=-4\pi G_N a^2\delta\rho,\qquad k^2(\phi'+\mathcal H\psi)=4\pi G_N a^2(\bar\rho+\bar P)\theta,$$

$$\phi'' + \mathcal{H}(\psi' + 2\phi') + (2\mathcal{H}' + \mathcal{H}^2)\psi + \frac{k^2}{3}(\phi - \psi) = 4\pi G_N a^2 \delta P, \qquad k^2(\phi - \psi) = 12\pi G_N a^2 (\bar\rho + \bar P)\sigma,$$

with the cold dark matter $\delta_c' = -\theta_c + 3\phi'$, $\theta_c' = -\mathcal{H}\theta_c + k^2\psi$; the baryons and photons coupled by Thomson scattering (n_e the free-electron density, σ_T the Thomson cross-section, $R = 4\bar\rho_\gamma/(3\bar\rho_b)$, c_s the baryon sound speed, $F_{\gamma l}$ and $G_{\gamma l}$ the temperature and polarisation multipoles, $F_{\gamma 2} = 2\sigma_\gamma$),

$$\delta_b' = -\theta_b + 3\phi', \qquad \theta_b' = -\mathcal{H}\theta_b + c_s^2 k^2 \delta_b + R\, a n_e \sigma_T(\theta_\gamma - \theta_b) + k^2\psi,$$

$$\delta_\gamma' = -\tfrac{4}{3}\theta_\gamma + 4\phi', \qquad \theta_\gamma' = k^2\big(\tfrac{1}{4}\delta_\gamma - \sigma_\gamma\big) + k^2\psi + a n_e \sigma_T(\theta_b - \theta_\gamma),$$

$$F_{\gamma l}' = \frac{k}{2l+1}\big[l F_{\gamma(l-1)} - (l+1)F_{\gamma(l+1)}\big] - a n_e \sigma_T\Big[F_{\gamma l} - \frac{\delta_{l2}}{10}\big(F_{\gamma 2} + G_{\gamma 0} + G_{\gamma 2}\big)\Big] \quad (l \ge 2),$$

$$G_{\gamma l}' = \frac{k}{2l+1}\big[l G_{\gamma(l-1)} - (l+1)G_{\gamma(l+1)}\big] + a n_e \sigma_T\Big[-G_{\gamma l} + \frac{1}{2}\big(F_{\gamma 2} + G_{\gamma 0} + G_{\gamma 2}\big)\Big(\delta_{l0} + \frac{\delta_{l2}}{5}\Big)\Big];$$

and one massive hierarchy per neutrino mass state i (comoving momentum q, energy $E_i = \sqrt{q^2 + a^2 m_i^2}$, multipoles $\mathcal{N}_l^{(i)}$ of the perturbed Fermi-Dirac distribution $\bar f_\nu$): $\mathcal{N}_0^{(i)\prime} = -\frac{qk}{E_i}\mathcal{N}_1^{(i)} - \phi'\frac{d\ln \bar f_\nu}{d\ln q}$, $\mathcal{N}_1^{(i)\prime} = \frac{qk}{3E_i}(\mathcal{N}_0^{(i)} - 2\mathcal{N}_2^{(i)}) - \frac{E_i k}{3q}\psi\frac{d\ln \bar f_\nu}{d\ln q}$, $\mathcal{N}_l^{(i)\prime} = \frac{qk}{(2l+1)E_i}[l\mathcal{N}_{l-1}^{(i)} - (l+1)\mathcal{N}_{l+1}^{(i)}]$ ($l \ge 2$). CAMB 2.0.5 solves them with the Planck 2018 parameters above, θ_{MC} fixed (so H_0 is an output), $\ell_{max} = 2600$, lensing included. UM-2 has three massive eigenstates, each with the degeneracy $N_{eff}/3 = 1.0144$; CAMB gives each the density deg $\times \rho_{FD}(m'; T_\nu)$, so the masses passed are $m_i' = m_i \deg^{-1/4}$; because CAMB recovers them by inverting the relativistic density from ω_ν and the mass fractions, ω_ν and the fractions are iterated (damped exponent 0.8, 13 iterations) until the masses agree to 7.8×10^{-12}. The comparison model is the CAMB default (one neutrino of 0.06 eV, $N_{eff} = 3.044$).

- **Relic abundance.** For $\mathcal{L} = G\,\bar\chi\chi F_{\mu\nu}F^{\mu\nu}$ per abelian channel, $\sigma v = G^2 s(s - 4M^2)/(4\pi) \to G^2 M^4 v^2/\pi$ (p-wave); in total $\sigma v = (M^4 v^2/\pi)(8G_g^2 + 3G_W^2 + G_B^2)$, $\langle\sigma v\rangle(x) = 6b/x$ with $b = (M^4/\pi)(8G_g^2 + 3G_W^2 + G_B^2)$; freeze-out (Kolb and Turner, Dirac fermion, $g = 2$, $c = 1/2$): $x_f = \ln\big[0.038\, c(c+2)\, g\, M_{Pl} M \langle\sigma v\rangle(x_f)/\sqrt{g_* x_f}\big]$, $\Omega h^2 = 2 \times 1.07 \times 10^9\ \mathrm{GeV}^{-1}\, x_f/\big(\sqrt{g_*}\, M_{Pl}\, 3b/x_f\big) = 0.060$ per species (two species), with $M_{Pl} = 1.22091 \times 10^{19}$ GeV and $g_* = 105.75$ (three generations without ν_R, the gauge bosons; no Higgs; the fold bands not thermalised). Unitarity: the operator annihilates a pair only from ${}^3P_0$ ($J = 0$), so for unpolarised Dirac fermions $\sigma \le \pi/(4p^2)$, $p = Mv/2$, i.e. $\sigma v \le \pi/(M^2 v)$, whose thermal average over $f(v) \propto v^2 e^{-xv^2/4}$ is $\langle\pi/(M^2 v)\rangle = \sqrt{\pi x}/M^2$ (checked by numerical averaging); the effective cross-section $G^2 M^4 v^2/\pi$, $G = (8G_g^2 + 3G_W^2 + G_B^2)^{1/2}$, exceeds the bound for $v > v_c = (\pi/(GM^3))^{2/3}$.

- **Direct detection** (ESTIMATE). Trace anomaly: $\langle N|G^\alpha G^\alpha|N\rangle = -\frac{8\pi}{9\alpha_s} m_N f_{TG}$, $f_N = G_g \frac{8\pi}{9\alpha_s} m_N f_{TG}$, $\sigma_{SI} = \mu_N^2 f_N^2/\pi$ (μ_N the dark-matter-nucleon reduced mass; $f_{TG} = 0.92$, $\alpha_s = 0.3$, $m_N = 0.939$ GeV, $G_g = G/\sqrt{8}$ if the whole coupling were in gluons), against the LZ limit scaled linearly from 2.2×10^{-48} cm^2 at 40 GeV.

- **Condensate.** A homogeneous condensate $S = \bar\Psi\Psi \propto a^{-3}$ with $U = \frac{\lambda}{2}S^2$ has $\rho = mS + U$, $p = S\,dU/dS - U = \frac{\lambda}{2}S^2$, so

$$w(a) = \frac{X(a)}{1 + X(a)}, \qquad X(a) = \frac{\lambda S}{2m} = X_0 a^{-3}, \qquad X_0 = \frac{w_0}{1 - w_0}, \qquad w_a \equiv -\frac{dw}{da}\Big|_{a=1} = \frac{3X_0}{(1 + X_0)^2},$$

and the energy density $\rho = mS(1 + X)$ crosses zero at $a = (-X_0)^{1/3}$ when $\lambda < 0$.

- **Uncertainty relation.** For bosons, with $u = e^{-3\eta\xi/2}\tilde u$: $e^{3\eta\xi}u'^2 = \tilde u'^2 + \frac{9}{4}\eta^2\tilde u^2 - \frac{3}{2}\eta(\tilde u^2)'$, hence $(m\,fL)^2 = \langle p^2\rangle + \frac{9\eta^2}{4} - \frac{3\eta}{2}\varrho(\mathrm{fold}) + (\frac{3\eta}{2} + \beta)\varrho(\mathrm{tip})$ with $\langle p^2\rangle = \int \tilde u'^2/\int \tilde u^2$ and the normalised density $\varrho = \tilde u^2/\int \tilde u^2$ ($\varrho = |\chi|^2/\int|\chi|^2$ and $\langle p^2\rangle = \int|\chi'|^2/\int|\chi|^2$ for fermions); the Heisenberg estimate is $m_H = \hbar c/(2\Delta s)$ with $\Delta s = fL\,\Delta\xi$, $\Delta\xi^2 = \int \xi^2 \varrho\, d\xi/\int \varrho\, d\xi$ the mean square distance from the fold. On the cell the relation reads $\sqrt{\langle p^2\rangle}\,\Delta\xi \ge (1 - \varrho(\mathrm{tip}))/2$, from $1 = \varrho(\mathrm{tip}) - \int \xi\varrho'\, d\xi$ and the Cauchy-Schwarz inequality (a theorem); it is also checked numerically for each of the 21 tabulated states (five boson modes, two doublet and twelve singlet zero modes, two states of Ψ).

- **$W^\pm$, Z fit.** $\rho(\beta_W) - 1 = \Lambda_W(\beta_W)/[\Lambda_Z(\beta_W/\cos^2\theta_W)\cos^2\theta_W] - 1$ with $\cos^2\theta_W = (m_W^2/m_Z^2)/\rho_0$ is solved for $\rho - 1 = \rho_0 - 1$ by bisection in $\ln\beta_W$ (80 steps); the model's $m_Z = m_W\sqrt{\Lambda_Z/\Lambda_W}$ then equals the measured value to 10^{-8} (checked).

- **Cosmic variance.** The relative error of a single multipole is $\sqrt{2/(2\ell+1)}$; the binned full-sky $\chi^2 = \sum_b (\bar\delta_b/\sigma_b)^2$ uses bins of $\Delta\ell = 50$, the mean relative deviation $\bar\delta_b$ and $\sigma_b = \sqrt{2/\sum_{\ell\in b}(2\ell+1)}$.

- **Thermal history (version 3).** For every species i in equilibrium (the x_0 masses; no Higgs particle, no right-handed neutrinos, no fold bands; the dark matter is not counted, as it would add less than 0.1 at 5 TeV) the ideal-gas integrals

$$\rho_i = \frac{g_i}{2\pi^2}\int_0^\infty \frac{p^2 E\,dp}{e^{E/T}\pm 1}, \qquad p_i = \frac{g_i}{6\pi^2}\int_0^\infty \frac{p^4\,dp}{E(e^{E/T}\pm 1)}, \qquad s_i = \frac{\rho_i + p_i}{T},$$

  evaluated by 160-point Gauss-Legendre quadrature in $u = p/T \le [(m/T+60)^2 - (m/T)^2]^{1/2}$, give $g_\rho = 30\sum \rho_i/(\pi^2 T^4)$ and $g_s = 45\sum s_i/(2\pi^2 T^3)$ on 360 temperatures from $10^{-4.5}$ to $10^{4.3}$ GeV; the quark-gluon plasma carries the weight $w = \frac12[1+\tanh((T-0.155\text{ GeV})/0.015\text{ GeV})]$ and the hadron gas $1-w$; below 2 MeV $T_\nu/T = (g_{s,e\gamma}/(11/2))^{1/3}$.

- **Relic abundance from the Boltzmann equation (version 3).** For one Dirac species (Y total, $x = M/T$)

$$\frac{dY}{dx} = -\frac{(1+\delta)s\langle\sigma v\rangle}{2xH}(Y^2 - Y_{eq}^2), \qquad \delta = \frac13\frac{d\ln g_s}{d\ln T}, \qquad H = \Big(\frac{8\pi^3 g_\rho}{90}\Big)^{1/2}\frac{T^2}{M_{Pl}}, \qquad Y_{eq} = \frac{4M^2 T K_2(x)}{2\pi^2 s},$$

  integrated for $\ln Y$ with the implicit Radau method (relative tolerance 10^{-10}) from $x = 5$ to 2000; $\Omega h^2 = MY_\infty s_0/(\rho_c/h^2)$. The thermal average

$$\langle\sigma v\rangle = \frac{1}{8M^4 T K_2(x)^2}\int_{4M^2}^\infty \sigma(s)(s-4M^2)\sqrt{s}\,K_1(\sqrt{s}/T)\,ds$$

  is computed by adaptive quadrature in the variable $s/(4M^2) - 1$ (the kinetic energy of the pair in units of its rest energy) with exponentially scaled Bessel functions and break points of this variable at $1/x$, $5/x$, $20/x$ and at any unitarity crossing, tabulated on 200 to 260 points in x and interpolated by a cubic spline in $(\ln x, \ln\langle\sigma v\rangle)$; the fraction of the annihilation rate from collisions above a crossing s_c is the same integral restricted to $s > s_c$ over the full integral (adaptive quadrature up to $200/x$ in this variable with the same break points, relative tolerance 10^{-11}), and the exact crossing is quoted by its relative velocity $v = 2\beta$; the strengths for $\Omega h^2 = 0.060$ per species are found by Brent's method in $\ln G$. The cross-sections and bounds: $\sigma_{I12} = G^2 s^{3/2}(s-4M^2)^{1/2}/(8\pi) \le \pi/(4p^2)$ ($^3P_0$) and $\sigma_{I13} = \sum_f N_c G_f^2 s(1+2M^2/s)/(24\pi\beta) \le 3\pi/(4p^2)$ ($^3S_1$), $G_f = G_A\frac13\tilde I_f^{(b)}$, $\beta = (1-4M^2/s)^{1/2}$, $p^2 = s/4 - M^2$; a capped cross-section is $\min(\sigma, \text{bound})$. Validation: an s-wave 4.4×10^{-26} cm^3/s against the freeze-out formula and the late-time integral $1/Y_\infty - 1/Y(x_*) = \int_{x_*}^\infty (1+\delta)s\langle\sigma v\rangle/(2xH)dx$.

- **Freeze-in of the fold bands (version 3).** The pair-production rate density

$$\gamma = \frac{T}{32\pi^4}\int_0^\infty ds\,\Big(\sum_{spins}\sigma\Big)s^{3/2}K_1(\sqrt{s}/T) = \frac{4}{\pi^5}N_\chi G_V^2 T^8\sum_f N_c(q_f\tilde I_f)^2$$

  follows with $\int_0^\infty x^6 K_1(x)dx = 384$ (checked by quadrature, as is $\int_0^\infty x^7 K_2(x)dx = 3072$, the energy moment that gives $8T$ per pair); $d(\rho_{fold}a^4)/d\ln(1/T) = 8T\gamma a^4/H$ with $a = g_s^{-1/3}/T$, integrated by the trapezoidal rule on 600 logarithmic temperatures from T_{max} to 1 GeV; $\Delta N_{eff} = \rho_{fold}/(\frac78\frac{\pi^2}{15}T_\nu^4)$ today.

- **Right-handed neutrinos for the $B-L$ alternative (version 3).** Per flavour, $dn_{\nu_R}/dx_4 + 3Hn_{\nu_R} = \gamma_\nu(1 - n_{\nu_R}^2/n_{\nu_R,eq}^2)$ with $\gamma_\nu = \sum_{A,B} n_{\chi,eq}^2 G_\nu^2 M^2/(2\pi)$, $n_{\chi,eq} = 2(MT/2\pi)^{3/2}e^{-x}$, $n_{\nu_R,eq} = \frac34\zeta(3)T^3/\pi^2$, $G_\nu = G_A^{B-L}\tilde I_{\nu_R}^{(b)}$ (root mean square of the three) with $G_A^{B-L} = G_A[S_B/S_{B-L}]^{1/2}$, $S = \sum_f N_c(q_f\tilde I_f^{(b)})^2$, integrated (Radau) for $Y = n/s$ from $x = 2$ with $Y = 0$; the elastic rate $\Gamma_{el} = 4n_{\chi,eq}\sigma_{el}$, $\sigma_{el} = G_\nu^2(3T)^2/(2\pi)$ for a Weyl ν_R on a heavy Dirac fermion (ESTIMATE); decoupling where $\gamma_\nu/(n_{\nu_R,eq}H)$ and Γ_{el}/H fall below 1 on a grid of step 0.01 in x; $\Delta N_{eff} = 3(10.75/g_s)^{4/3}F_h$. With J^B the production through I11 is compared with the I13 annihilation by the ratio $G_V^2\sum_i \tilde I_{\nu_R i}^2/(G_A^2 S_B)$.

- **The self-interaction channel (version 3, after the review).** $\tilde I_\lambda = \int_{-L}^0 s_{DF}^2 e^{-3Hy}dy$ with $s_{DF} = a_D b_F + b_D a_F$ (quadrature of the free orbitals), $G_\lambda = \lambda H\tilde I_\lambda/(2f^4 v_t)$, $\sigma v = G_\lambda^2 M^2/(2\pi)$ per fold species (ESTIMATE), $\gamma_\lambda = \sum_{A,B} n_{\chi,eq}^2\sigma v N_F$; the energy $d(\rho_{fold}a^4)/d\ln(1/T) = 2M\gamma_\lambda a^4/H$ is integrated by

the trapezoidal rule on 800 logarithmic temperatures from T_{max} to the freeze-out; the fold bands are in equilibrium where $2\gamma_\lambda/(n_{fold,eq}H) \geq 1$ (600 temperatures), and then decouple as four Weyl species, $\Delta N_{eff} = 4(10.75/g_s)^{4/3}F_h$, at the lowest such temperature.

- **The algebra of I13 (version 3, after the review).** With the author's 16×16 matrices (exact sympy): $(iC\gamma^M P_b)^\dagger - iC\gamma^M P_b = 0$ for $M = x_1,\dots,x_7$ and $\neq 0$ for $M = x_0$, where it equals $iC\gamma^{x_0}P_a - iC\gamma^{x_0}P_b$; $CP_bC^{-1} = P_a$; $[P_b, \gamma^\mu\gamma^\nu] = 0$; $[B, C\gamma^M P_b] = 0$ for $M = x_1,\dots,x_4$, $\{B, C\gamma^M P_b\} = 0$ for $M = x_5, x_6, x_7$; $QC\gamma^M Q = 0$ for $M = x_5, x_6, x_7$ with $Q = (1 \mp B)/2$.

- **The coupled solution in environments (version 3).** The overlaps are quadratures of the free orbitals (80 functions per sector, 800 quadrature points) with the zero-mode profiles; their checks are self-consistent Kohn-Sham states with the mean fields $v = e\varrho_f e^{-3Hy}$ or $v_b = e\varrho_f e^{-3Hy}$ (the latter on the lower component only, $\frac{1}{2}(1-\sigma_3)v_b$) at $e = \pm10^{-4}, \pm2\times10^{-4}$, the derivative from $(4D_1 - D_2)/3$ with D_1, D_2 the central differences. The mean fields of an environment are $\delta\varepsilon = \sum_f (G_V q_f \tilde{I}_f + G_A q_f^B \tilde{I}_f^{(b)})n_f + \mathcal{O}\frac{c}{\Lambda_\star^3}[\langle g_s^2 G^2\rangle + \langle g^2 W^a W^a\rangle + \langle g'^2 B^2\rangle]$ with each chirality carrying half of an unpolarised density, $\langle\frac{\alpha_s}{\pi}G^2\rangle_T = \langle\frac{\alpha_s}{\pi}G^2\rangle_0 - \frac{8}{b}(\rho-3p)_{QCD}$ ($b = 11 - \frac{2}{3}n_f$), $(\rho-3p)_{QCD}/T^4 = 4e^{-\ln^2(T/0.2\,\mathrm{GeV})/0.5}$ for $0.05 < T < 1$ GeV and $\frac{b}{3}(1+\frac{5n_f}{12})\alpha_s^2$ above (α_s at $2\pi T$, one loop), and for $W^\pm$, Z the free part $\langle F^2\rangle_T = -2(\rho-3p)_V$ plus the order-α interaction part $\langle F_{geo}^2\rangle = 4g^2 p_2$, $p_2 = -(g^2T^4/144)d_A[C_A + \frac{5}{2}T_F N_f]$ (Feynman-Hellmann; the couplings at $2\pi T$ from the $\overline{MS}$ values at m_Z; below m_W times $\rho(m_W,T)/\rho(0,T)$), i.e. $\langle g^2 W^a W^a\rangle = g^2[2(-2I_W) + \cos^2\theta_W(-2I_Z)] + 4g^2 p_2^{SU(2)}$, $\langle g'^2 B^2\rangle = g'^2\sin^2\theta_W(-2I_Z) + 4g'^2 p_2^{U(1)}$; the early-Universe environments have the net baryon density $\eta_b n_\gamma$ ($n_\gamma = 2\zeta(3)T^3/\pi^2$) split as $n_n/n_p = e^{-Q/T}$, the dark matter in equilibrium for $M/T < 40$, and the freeze-out environment $T = M/x_f$ with the version-3 $x_f = 31.23$. The block response is the full adiabatic response $\delta k = (c_\rho\kappa\delta\rho + c_3\kappa\delta p_3 + c_p\kappa\delta\bar{p})/H^2$ ($c_\rho = 0.035852$, $c_3 = -0.107556$, $c_p = -0.140028$ from `background.json`) to the block averages $\kappa\delta X = 8\pi G_N G(k_*)X_4$ of all species, with $\bar{p} = \frac{1}{4}\partial p/\partial\ln f$: $-\frac{1}{4}(\rho-3p)\,d\ln m/d\ln f$ for the plasma species (ideal gases with their x_0 masses), the net nucleons and electrons (degenerate Fermi gases at $T = 0$) and the dark matter, $-\frac{1}{4}(\rho-3p)_{QCD}\,d\ln\Lambda^{(n_f)}/d\ln f$ for the QCD plasma and $\frac{1}{4}p_2^{(i)}\,d\ln\alpha_i/d\ln f$, $d\ln\alpha_i/d\ln f = -\alpha_i[4/\alpha_i(m_{KK}) + b_i/(2\pi)]$, for the electroweak plasma (whose trace $-\frac{b_i\alpha_i}{2\pi}p_2^{(i)}$ enters $\rho - 3p_3$ with the QCD interaction measure); the loop gain is the central difference of $\delta\ln f$ in a trial $\delta\ln f = \pm10^{-3}$ of all masses, couplings and Λ_{QCD}; iterated with $\delta\ln m_s = (d\ln m_s/d\ln f)\delta\ln f$, $\delta\ln f = -\delta k/(2k_*)$, until the change is below 10^{-12} relative; the nucleon sensitivity is $d\ln m_p/d\ln f = (1 - \frac{\sigma_{\pi N}+\sigma_s}{m_p})[\frac{21}{27}(-1 - \frac{8\pi}{7\alpha_s(m_{KK})}) + \frac{2}{27}\sum_{c,b,t}\frac{d\ln m_q}{d\ln f}] + \frac{\sigma_{\pi N}}{2m_p}(\frac{d\ln m_u}{d\ln f} + \frac{d\ln m_d}{d\ln f}) + \frac{\sigma_s}{m_p}\frac{d\ln m_s}{d\ln f}$ with $1/\alpha_s(m_{KK}) = 1/\alpha_s(m_Z) + \frac{7}{2\pi}\ln(m_{KK}/m_Z)$. The W problem with the kinetic function $K = 1 + e\,s_{DM}$, $-(Ke^{3\eta\xi}u')'/(Ke^{3\eta\xi}) = \Lambda u$, $u'(0) = 0$, $K(-1)u'(-1) = \beta_W u(-1)$, is solved with 300 quadratic finite elements at $e = \pm10^{-3}$, $\pm2\times10^{-3}$ (Richardson), and with $\beta = 0$ for the photon.

- **The metric series (version 3).** The graviton tower $u_n = e^{-a\xi}(\cos n\pi\xi + \frac{a}{n\pi}\sin n\pi\xi)$, $\Lambda_n = a^2 + n^2\pi^2$, $a = 3\eta/2$, is checked against 400 quadratic finite elements with natural boundary conditions; the norms $N_n = \frac{1}{2}(1 + a^2/q_n^2)$, the overlaps $\int\varrho u_n\,d\xi$ and the orthogonality $\int e^{3\eta\xi}u_n\,d\xi = 0$ by the trapezoidal rule on 200001 points; $\alpha_n = (N_0/N_n)(\int\varrho_M u_n)(\int\varrho_m u_n)$, for two masses with the same profile $(N_0/N_n)(\int\varrho u_n)^2$. The potentials come from a second CAMB run of each model without the CMB (so that the CMB numbers are unchanged), with the matter power spectrum to $k = 20$ Mpc^{-1} at $z = 2$, 1, 0.5, 0: $\Delta_\Phi = [k^3 P_W(k)/(2\pi^2)]^{1/2}/k^2$ from the Weyl variable $k^2(\phi+\psi)/2$, against $\frac{3}{2}\Omega_m(H_0/c)^2[k^3P_\delta/(2\pi^2)]^{1/2}/k^2$ from the total density contrast, and against $\frac{3}{5}A_s^{1/2}(k/k_0)^{(n_s-1)/2}g(0)$ with $g(0) = \frac{5}{2}\Omega_m E(1)\int_0^1 db/[bE(b)]^3$; σ_8 and $f\sigma_8$ are CAMB's.

- **Majorana alternative (version 3).** $\kappa_{Mi} = m_i f L/\zeta(c_L)^2$; $m_{\beta\beta}$ between $\max(0, 2\max_i t_i - \sum_i t_i)$ and $\sum_i t_i$, $t_i = |U_{ei}|^2 m_i$, confirmed by a 721×721 scan of the two physical Majorana phases.

8.3 Corrections made during the work

1. The gauge-invariance test of the tip terms first used the wrong sign of the hypercharge phase; with $\psi \to e^{iY\theta}\psi$ every term is invariant to 4×10^{-15}.
2. The finite-element cross-check of Λ_W was first limited by round-off; the gradient-form Rayleigh quotient with Richardson extrapolation (observed order 4.01) agrees with the exact value to 3×10^{-14}.
3. The Kohn-Sham basis convergence and the conservation law were first checked without extrapolation; Richardson in the basis size (observed orders 4.95 for the energy, 2.97 for the hidden pressure) was added.

4. The self-consistency loop did not converge for $N = 592$ at the record's coupling; a fixed labelling of the occupied set and a safeguarded Anderson mixing were introduced, the benchmark coupling λ_B was adopted, and the record's coupling is kept as a stress test, which now converges.
5. A first statement that every primordial start above the unstable static block reaches the stable one was false in detail; the separatrix $k_c(h_0)$ was computed by bisection instead.
6. The primordial figure was first sampled too coarsely (aliasing of the radion oscillation) and is now densely sampled.
7. The sum rules of the uncertainty analysis were first imprecise for the sharp fermion zero modes; exact mpmath normalisation and the tip term fixed them (10^{-8}).
8. The radion quantity was stored under a wrong sign label; it is now stored as $\omega_r^2 =$ det of the block Jacobian.
9. Portal names of the first universe model (P1 to P6) in two check texts were replaced by I9, I11, I12.
10. The gauge-kinetic portal was first written with the scale to the power 5, which is dimensionally wrong in the geometric normalisation of the gauge fields; it is $\bar{\Psi}\Psi\, F_{MN}F^{MN}/\Lambda_\star^3$, and the symbol of the portal scale was changed so that it no longer clashes with the Dirichlet bound Λ_D.
11. Before publication every number of the documents was compared with the computed outputs; four statements were corrected (the dark-matter momentum $p\,fL = 1.89901$, the finite-element agreement 5×10^{-10}, the TT and EE agreements of version 1, and the restriction $A = 0$ of the sourceless static block).
12. The kinetic term of the tip sigma field was first written with the sign of the signature $(+,-,-,-)$; in the signature $(+,+,+,-)$ of $x_1,\dots,x_4$ it is $-\frac{v_\Sigma^2}{4}\mathrm{tr}((D_\mu\Sigma)^\dagger D^\mu\Sigma)$, which gives the mass terms $-\frac{1}{2}m^2V_\mu V^\mu$ with $m^2 > 0$ (the computations always used $m^2 > 0$).
13. With the real matrices $C\gamma^a$ antisymmetric, $\bar{\Psi}\gamma^a\Psi$ is anti-Hermitian; the portal currents are now written as the Hermitian currents $i\bar{\Psi}\gamma^M\Psi$ and the Standard-Model currents as $i\bar{\psi}\gamma^M\psi$.
14. The block response to the hidden pressure of a Kohn-Sham state first used p_0 alone; the common block scale is driven by the block-averaged pressure $\bar{p} = (p_0 + 3p_t)/4$, and $p_0 - p_t$ drives a block-shape mode outside the order-0 metric (OPEN). The linearised sourced equations were added (`um2_background.py`, 2 checks); they first omitted the density, which enters at first order through $3G(k_*)A^2 = \kappa\delta\rho$; with it the adiabatic response reproduces the four-dimensional Friedmann acceleration equation exactly.
15. The statement that N_A and N_B are separately conserved was first recorded as an asserted check; it is now computed with the author's matrices (commutators of B with C, γ^a, $C\gamma^a$, S^{ab}), which also shows that extra-time momenta, outside the good sector, would mix the halves.
16. The radion lifetime was first quoted as an unverified estimate; it is now computed (ESTIMATE, with the coupling factor C_r explicit), as is the bound on the block rate today.
17. Figures: the BAO legend showed the data label twice and two labels of the uncertainty figure overlapped; the condensate figure no longer draws the vertical lines where w diverges.
18. An independent review of the five documents of version 1 against the outputs found, and the documents were corrected: the first-order fermion mass formula (it needs $\zeta(-c_R)$ for the singlet, $c_R < 0$; the code always used $|c|$); the gap statement, which holds for the boson operators but not for Ψ (the dark matter, 44.35 TeV, lies below the boson gap of 99.5 TeV); the sensitivity of G_N to the block, which is $d\ln G_N/d\ln f = -4 + 2k_*G'(k_*)/G(k_*) = +2.05$ and not -4 (recomputed; the lunar-ranging bound becomes $7.1 \times 10^{-4}H_0$); ΔN_{eff} of a thermalised fold band (recomputed); a level spacing (0.25000); a rounding (1.948); the definitions of M_{Pl}, M_{Pl}, the radion formulas, Δs, the unitarity estimate and the cosmic-variance measure; the statement on proper and coordinate localisation; and several notational clashes ($\kappa = e^{-a_4}$ is now written out).
19. A second review of version 1 found, and the documents were corrected: the electroweak parameter was written $\rho = m_W^2/(m_Z^2\cos^2\theta_W)$ next to the on-shell definition $\cos^2\theta_W = m_W^2/m_Z^2$, which makes it identically 1; the computation always used $\rho = \Lambda_W/(\Lambda_Z\cos^2\theta_W)$. Checking the uncertainty relation for every state then showed that the bound $\langle p^2\rangle \geq 1/(4\Delta\xi^2)$ quoted for the hidden kinetic energy is not valid on the cell: the tip, where the cell ends, adds a boundary term, and the valid relation is $\sqrt{\langle p^2\rangle}\,\Delta\xi \geq (1 - \varrho(\mathrm{tip}))/2$ (the broad fold zero mode of Ψ has $\langle p^2\rangle\Delta\xi^2 = 0.097$); a computed check of it was added. Further: the $\overline{MS}$ angle in the Z widths, the shooting accuracy, the normalisation volume Vol_7 (not the proper seven-volume), the occupations f_o, the coupling matrix Λ_ℓ, and the notation (with a list of the symbols with more than one use in each document).
20. A third review of version 1 found, and the computations and documents were corrected: (a) the $W^\pm$, Z fit set the angle of the gauge couplings equal to the on-shell angle of the measured masses and fitted the tree-level $\rho - 1$ to $\rho_0 - 1$, which put the model's $m_Z = 91.1706$ GeV 17 MeV below the measured value; since ρ_0 measures the custodial violation beyond the Standard-Model loops, the angle of the couplings is now fixed by the measured masses, $\cos^2\theta_W = (m_W^2/m_Z^2)/\rho_0 = 0.776368$, and the model reproduces both masses

(check `mZ_reproduced`); $\beta_W = 0.011912$ and every scale in GeV moved by $+8.50 \times 10^{-4}$ ($m_{KK} = 121.32$ TeV, $H = 14.296$ TeV, gap 99.48 TeV, radion 48.42 TeV, $M_{DM} = 44.35$ TeV). The tree-level Z widths with the angle of the couplings were added (check `Z_width_scheme`). (b) The familiar bound $\langle p^2 \rangle \geq 1/(4\Delta\xi^2)$ is guaranteed only for profiles negligible at the tip. (c) Symbol lists and wording.

21. **Version 2 (2026-10-18): three generations and three neutrino flavours.** By the author's orders of 2026-10-18 (option A):
    (a) three generations of quarks and leptons with one doublet bulk mass per sector, the twelve masses as exact x_0 eigenvalues and the PMNS and CKM matrices as the tip matrices, proved exact for three generations (`um2_flavour.py`, new; `um2_spectra.py`, `um2_fields.py` extended); (b) the flavour conversion of the three neutrinos (`um2_neutrino.py`, new): vacuum, matter, accelerators, the Sun, the early Universe and N_{eff}, the CNB of three mass states, supernovae; (c) the chiral tip Z portal I9 is removed: with three neutrinos it has no role left, its even part is bounded by the invisible width ($N_\nu = 3 + 2r^2$, $r^2 < 0.0056$) and would still keep the fold band in equilibrium to 14.7 MeV ($\Delta N_{eff} = 2.02$), and its oriented part would thermalise the fold band through the decays of the odd Ψ_A states ($\Delta N_{eff} = 0.0958$ to 0.1257, 2.1 to 2.4 standard deviations above ACT DR6); consequently both halves of Ψ keep an exact fold parity and the dark matter consists of the lightest odd states of both halves (two species, relic recomputed: $x_f = 30.38$, $(15.5\ \text{TeV})^{-3}$); (d) the invisible width uses the PDG 2025 value $N_\nu = 2.9963 \pm 0.0074$ instead of the LEP 2006 value; (e) N_{eff} is computed for three flavours (3.0476 to 3.0485) and the precision value 3.0432 (Cielo et al.) replaces 3.044; (f) the cosmology has three massive neutrinos ($\sum m_\nu = 0.0599$ eV) instead of one of 0.05 eV and no dark radiation; (g) the Z widths of three generations (2485.2 MeV with the QCD factor) remove the one-generation conflict (1232 MeV); (h) the Kohn-Sham portal sources use the three-generation localisations. Errors found and corrected during the version-2 work: the 1-3 MSW resonance energy was first evaluated with a spurious factor $1/c_{13}^2$ (11.49 GeV); it is $E = \Delta m^2_{ee} \cos 2\theta_{13}/(2V_{CC}) = 11.135$ GeV with $\Delta m^2_{ee} = c_{12}^2 \Delta m^2_{31} + s_{12}^2 \Delta m^2_{32}$, confirmed by the exact three-flavour root 11.133 GeV (now a check); CAMB's built-in normal-hierarchy option put the mass in one eigenstate, so the three eigenstates are now set up explicitly with the iteration described in the methods; a check of the analysis (fermions below the uncertainty estimate) first used a threshold that the muon and the strange quark exceed and now tests the documented statement ($m/m_H < 0.01$ for every fermion); in the documents the muon eigenvalue (4.7795×10^{-6}), the Landau-Zener bound ($P_{LZ} \leq 2.6 \times 10^{-77}$) and the explanation of $\Omega_\nu h^2$ (the non-relativistic coefficient 94.06 eV at the instantaneous-decoupling temperature, 93.14 eV with the heating) were corrected; the published values of the decoupling method were read from the source (Escudero 2020, Table 1) and the separate-temperature value 3.048 added to the validation.

22. **Independent review of version 2 (2026-10-18).** A review of the five documents of version 2 against the outputs found every computed number in agreement except three roundings, and found the following, all corrected in the documents: (a) the dark-matter relic: the operator $\bar{\chi}\chi F_{\mu\nu}F^{\mu\nu}$ annihilates a pair only from ${}^3P_0$, so the unitarity bound on the spin-averaged cross-section is $\sigma \leq \pi/(4p^2)$ and $\langle \sigma v \rangle \leq \sqrt{\pi x_f}/M^2 = 5.79 \times 10^{-26}$ cm^3/s; the freeze-out requirement 2.01×10^{-25} cm^3/s exceeds it 3.47 times (the documents had compared it with $(4\pi/M^2)\sqrt{x_f/\pi} = 2.32 \times 10^{-25}$ cm^3/s, the bound of one helicity channel without the spin average, and called the requirement 87 % of it); the relic abundance of the dark matter is therefore OPEN in version 2, and the check `B3_relic_requirement_and_unitarity` of `um2_analysis.py` now computes the bound, its numerical thermal average, the crossing velocity $v_c = 0.263$ and the fraction (79 %) of pairs above it; (b) "no tip operator reaches Ψ_B" holds only for chiral P_+ tip currents and for operators built on $\bar{\Psi}\Psi$ (a non-chiral tip current would reach Ψ_B); (c) the fast-flavour dispersion relation is written with the upper-index shifts Λ^0, Λ^3 and the stated metric $\eta^{\mu\nu} = \text{diag}(1, -1)$ on (x_4, x_3); (d) $c_Z = c'_Z = 0$ is a modelling choice (ASSUMED): the invisible width bounds the even part and N_{eff} disfavours, but does not exclude, the oriented part; (e) the coordinate rule: the flavour, decoupling and fast-flavour equations had written t for x_4 and z for the symmetry axis x_3; (f) roundings: the adiabatic solar P_{ee} at 15 MeV (0.306), the τ singlet bulk mass ($-9.205H$), the Kohn-Sham accuracy (2.2×10^{-9} relative); (g) minor items: J_{max} computed from the best-fit angles (0.0335) against NuFIT's marginalised value (0.0333 ± 0.0007); the conventions of the supernova resonances (two-flavour conditions); the tip matrices equal the mixing matrices only with $\kappa_e = \kappa_u = 1$ (ASSUMED); the three roles of the measured values; the sum rules of the uncertainty analysis (absolute 1.0×10^{-8}, relative 7.8×10^{-4} and 6.0×10^{-4} for $W^\pm$ and Z); the 50-digit roots against the stored masses (4.5×10^{-11}); the labels ESTIMATE and STANDARD; the number required for a single species ($(17.6\ \text{TeV})^{-3}$ with $g_* = 105.75$, not a version-1 value); further notational clashes added to the symbol lists; (h) self-containment: equations and inputs used but not written in a document (the relic and ΔN_{eff} formulas,

the Einstein-Boltzmann equations, the BAO χ^2 and data, the nonlinear background equations, the fermion equations with their boundary conditions, the decoupling equations, the CNB, Landau-Zener, adiabatic solar, condensate and pairing formulas, the Kohn-Sham equations behind the quoted equations of state) are now written out in every document that uses them.

23. **Version 3 (2026-10-18).** By the author's order for the coupled solution: (a) the computations `um2_thermal.py`, `um2_coupled.py` and `um2_series.py` (new), the matter power spectrum and the version-3 model in `um2_cosmology.py`, the Majorana alternative in `um2_neutrino.py`, and an optional potential on the lower component in the Kohn-Sham solver `um2_kohnsham.py` (the record's reference states reproduced again to 10^{-12}); (b) the portal I13 was first coupled to the $B-L$ current; the computation C9 then showed that $\chi\bar{\chi} \leftrightarrow \nu_R\bar{\nu}_R$ would thermalise the right-handed neutrinos and add $\Delta N_{eff} = 0.144$ (2.5 standard deviations above ACT DR6), so I13 couples to the baryon-number current (G_A 1.072 $\rightarrow$ 1.278×10^{-9} GeV^{-2}; xenon 0.12 $\rightarrow$ 0.16 of LZ); (c) the first version of the C9 rates compared the production rate of three flavours with the equilibrium density of one; corrected (per flavour) before the result was recorded; (d) `um2_thermal.py` read a key of the analysis output that item 22 had renamed; corrected; (e) the early-Universe environments first put the whole net baryon density into protons; it is now shared by weak equilibrium, $n_n/n_p = e^{-Q/T}$, and the QCD interaction measure had the wrong flavour thresholds in its perturbative branch and a special case at exactly 200 MeV (neither affected a stored number); (f) the text of the validity check C7 now refers to the benchmark strengths; (g) the thermal numerical result refines the version-2 statement on I12: a cross-section at the ${}^3P_0$ limit at every energy leaves $\Omega h^2 = 0.058$ per species, so I12 alone could give the relic only as a strongly coupled channel at the unitarity limit; (h) in MODEL a cross-reference to the neutrino decoupling pointed to section 9.4 instead of 10.4 (since version 2).

24. **Independent review of version 3 (2026-10-18).** A review of the five documents of version 3 at commit `726997f` against the outputs (2055 s, 74 tool calls) found every version-3 number in agreement with the stored outputs except the items below, and the following errors and omissions, all corrected in the computations and the documents (the affected scripts rerun; 235 checks, all passing):
    (a) the I13 bilinear $i\bar{\Psi}\gamma^M P_b \Psi$ is Hermitian only for $M \neq x_0$, since $(iC\gamma^{x_0}P_b)^\dagger = iC\gamma^{x_0}P_a$; the action now sums over $M = x_1, \ldots, x_7$ (in the frame $a = 1, \ldots, 7$), and the new check `C8_I13_algebra` verifies with the author's matrices the Hermiticity, its failure for x_0, $CP_bC^{-1} = P_a$, the Lorentz property and the B structure ($C\gamma^M P_b$ commutes with B for $x_1, \ldots, x_4$ and anticommutes for x_5, x_6, x_7, whose currents vanish for spinors in one half, so the Standard-Model zero modes, all in $B = -1$, are not mixed); no number changes, since the zero-mode reduction uses the components $M = \mu$ only; (b) the block response of the coupled solution kept only the hidden pressure of non-relativistic matter and only the $\delta\bar{p}$ term of the adiabatic response; it now uses the full response (c_ρ, $c_3 = -3c_\rho$, c_p) and the hidden pressure $\bar{p} = \frac{1}{4}\partial p/\partial \ln f$ of every species, the hot plasma included (ideal gases with the x_0 masses, the QCD interaction measure with $d\ln\Lambda/d\ln f$, the order-α electroweak pressure with $d\ln\alpha/d\ln f$): $\delta \ln f$ at the freeze-out $5.6 \times 10^{-45} \rightarrow 1.1 \times 10^{-34}$ (7×10^{-39} in the electroweak era, at most 9×10^{-49} from nucleosynthesis on), the loop gain $1.3 \times 10^{-44} \rightarrow 1.1 \times 10^{-33}$, the radion term of the metric series accordingly; the Kohn-Sham block feedback also reports the full response; the conclusion, a negligible back-reaction, is unchanged; (c) the thermal W, Z condensates had only the free massive-vector part; the order-α interaction part $\langle F^2_{geo}\rangle = 4g^2 p_2$ (Feynman-Hellmann; for QCD it reproduces the trace-anomaly condensate to 10^{-16}, now part of check C4) dominates above m_W: $\langle g^2 W^a W^a\rangle$ at the freeze-out -5.0×10^{11} GeV^4; (d) the freeze-out environment used the version-2 $x_f = 30.38$ ($T = 1.46$ TeV); it uses the version-3 $x_f = 31.23$ ($T = 1.42$ TeV), and $\delta\varepsilon/M$ there is -1.5×10^{-6} (was -1.6×10^{-6}); the degrees-of-freedom table lists 1.42 TeV instead of 1.46 TeV; (e) the thermal history did not state λ: the self-interaction contains the channel $\chi\bar{\chi} \rightarrow$ fold pairs through the transition scalar density $s_{DF} = a_D b_F + b_D a_F$; the new computation C10 gives $\Delta N_{eff} = 1.83 \times 10^{26} G_\lambda^2$ GeV^4 and the bound $G_\lambda \leq 2.3 \times 10^{-15}$ GeV^{-2}, now ASSUMED (the Kohn-Sham test value λ_B with $v_t = H^{-3}$ would thermalise the fold bands and add 0.19); (f) the elastic ν_R rate of the $B-L$ alternative used $\sigma = G^2E^2/\pi$ instead of $G^2E^2/(2\pi)$ for a Weyl ν_R on a heavy Dirac fermion: kinetic decoupling $M/T = 25.1 \rightarrow 24.4$ ($T = 1769 \rightarrow 1816$ GeV), $\Delta N_{eff} = 0.144$ unchanged;
    (b) the I11 production of ν_R, quoted as 10^{-14} of the I13 rate without a source, is now computed: 2.0×10^{-17}; (h) the fractions of the annihilation rate above the unitarity crossings were integrated inaccurately (one of them exceeded 1): they are now computed by adaptive quadrature with the break points of the thermal weight (I12 at the relic strength: 96 %; the I12 benchmark $5.3 \rightarrow 5.2 \times 10^{-4}$; I13 uncapped 24.5 $\rightarrow$ 24.8 %, capped 19.0 $\rightarrow$ 21.7 %); the exact I12 crossing at $\sqrt{s} = 89510$ GeV is quoted with its relative velocity $v = 2\beta = 0.269$ (0.274 is the non-relativistic formula); the ${}^3P_0$-limit relic compares the thermal average

and $\sqrt{\pi x}/M^2$ at the same x (5.33 against 5.84×10^{-26} cm^3/s); (i) stale or mixed statements: "$c_V = 0$" and "no dark radiation" or "no extra radiation" of version 2; the cosmology labels (UM-2 version 3 is the model with the freeze-in fold bands: $N_{eff} = 3.0442$, $\sigma_8 = 0.8118$, BAO $\chi^2 = 31.49$, $H_0 = 67.38$; the model without them is the neutrino content of version 2); "relic OPEN" qualified as the result of version 2; the check counts of `um2_neutrino.py` (16) and `um2_cosmology.py` (8); the amplitude of the kinetic-function shift at the freeze-out (1.7×10^{-16}, not below 10^{-50}); "perturbative" made precise (22 % of the freeze-out rate at the $J = 1$ bound; $G_f^{-1/2} = 30$ TeV for the quark doublet, below $\sqrt{s} \approx 89$ TeV); the ν_R couplings (the I11 coupling as well as the tip mixing); the dark matter is not counted in g_ρ, g_s (as in the code); roundings ($\tilde{I}^{(b)}_{\nu_R} = 0.301$ to 0.308, $g_\rho(200\ \text{GeV}) = 104.95$, $g(0) = 0.7878$, $\Delta_\Phi(1\ \text{Mpc}^{-1}) = 5.03 \times 10^{-8}$, the Poisson agreement 1.5×10^{-6}); the signature convention of the trace anomaly and of the Proca Lagrangian, written for $(+,-,-,-)$; the $\overline{MS}$ angle in the running couplings; α_n for two different profiles; the order-2 magnitude $(2.4 \times 10^{-5})^2 = 5.8 \times 10^{-10}$; a heading labelled DERIVED over estimates; (j) self-containment: the quark sensitivities $d \ln m_q / d \ln f$, the scale $2\pi T$ of α_s, the densities of the environments, the hadron list, Ω_m, the symbols N_c, N_χ, G_A^{B-L}, $\sum_{A,B}$, $g_{s,e\gamma}$, K_1, K_2, the Mandelstam s, β, J and the version-3 symbols are now stated in every document that uses them, and the variable $s/(4M^2) - 1$ is no longer called z (z is $6Hx_0$ or a redshift).

8.4 What is not in the repository, and why

By the author's instruction the following private inputs are never committed (they are git-ignored): the prompt files, the author's private PDF with the supernova dark-energy fits, the folders `dirac-main/` and `vendor/`, the archive `dirac-main_2.zip`, the files `Generalized_Kronecker_Delta.*`, and the notebook `WaveFunctionUniverse-4+4-Einstein-Lovelock.nb` with its `.mx` files. The paper "Neutrino Flavor Conversion.pdf" (Gogilashvili and Tamborra, arXiv:2605.16504) is a third-party publication and is not committed either; UM-2 cites its results. UM-2 uses from these inputs only what is written out in this document (the notebook formulas of the metric and of the gamma matrices, the two supernova fits, the explosion statistics).

8.5 Reproduction

From the repository root, with Python 3 and the packages sympy, numpy, scipy, mpmath, matplotlib and camb installed (`pip install sympy numpy scipy mpmath matplotlib camb`), the commands

```
python universe_model_2/code/um2_algebra.py
python universe_model_2/code/um2_geometry.py
python universe_model_2/code/um2_background.py
python universe_model_2/code/um2_fields.py
python universe_model_2/code/um2_spectra.py
python universe_model_2/code/um2_flavour.py
python universe_model_2/code/um2_neutrino.py
python universe_model_2/code/um2_ks_jobs.py validate
python universe_model_2/code/um2_ks_jobs.py spectrum
python universe_model_2/code/um2_ks_jobs.py ground8
python universe_model_2/code/um2_ks_jobs.py ground592
python universe_model_2/code/um2_ks_jobs.py darkmatter
python universe_model_2/code/um2_ks_jobs.py portals
python universe_model_2/code/um2_ks_jobs.py feedback
python universe_model_2/code/um2_electroweak.py
python universe_model_2/code/um2_primordial.py
python universe_model_2/code/um2_cosmology.py
python universe_model_2/code/um2_analysis.py
python universe_model_2/code/um2_thermal.py
python universe_model_2/code/um2_coupled.py
python universe_model_2/code/um2_series.py
python universe_model_2/tools/build_docs.py
```

rerun every computation in dependency order (each exits with code 0 if and only if all its checks pass) and rebuild the PDF documents from the Markdown sources (pandoc and a TeX Live installation with the `lmodern` package

are needed for the last command). Version 1 was rerun from fresh clones of the repository (2026-10-17 at commit `b3c3c1a`; 2026-10-18 at commits `fa7305f`, `9285bb6` and `5b73d16`), version 2 on 2026-10-18 at commit `4a6421d` (every computation as a separate step, 1027 s in all) and version 3 on 2026-10-18 at commit `726997f` (every computation as a separate step, 1100 s in all, the longest the primordial integration with 269 s): every check passed (205 of 205 for version 2, 233 of 233 for version 3) and every stored number was reproduced to a relative 10^{-9} or better (in both runs all numbers were identical except one Kohn-Sham integral, which differs by 1.1×10^{-12} relative, because the threaded linear algebra does not sum in a fixed order), and the LaTeX sources, the PDF documents and the figures were rebuilt byte-identical. Between the two runs, `um2_analysis.py` gained the ${}^3P_0$ unitarity bound of item 22 (new stored numbers, the same 23 checks). After the run of version 3 the fixes of its independent review (item 24) changed `um2_thermal.py`, `um2_coupled.py`, `um2_series.py`, `um2_cosmology.py` and the `feedback` job of `um2_ks_jobs.py`, which were rerun in dependency order (thermal, coupled, series; cosmology and feedback; every check passed, 235 in all). The document build is deterministic: no dates in the LaTeX sources, PDF dates and trailer identifiers suppressed, a fixed `SOURCE_DATE_EPOCH`.

Software used: Python 3.13.16, sympy 1.14.0, numpy 2.5.3, scipy 1.18.1, mpmath 1.3.0, matplotlib 3.11.2, camb 2.0.5, pandoc 3.1.3, pdfTeX 1.40.25 (TeX Live 2023).

8.6 History in the repository

UM-2 was started on top of commit `0cb4916`. Version 1: its computations were committed in the order algebra and geometry (`ca523d9`), background gravity (`0b3e10d`), field reductions (`6fc35a0`), mass spectra (`dcfd528`), Kohn-Sham (`2f3018e`), electroweak sector (`2e4c48d`), primordial era (`a502e62`), cosmology (`717b46e`), analysis (`a06b55e`), followed by the corrections listed above and the documents MODEL, PROVENANCE, FIELD_EQUATIONS, SOLUTION and ANALYSIS (Markdown, LaTeX and PDF each); the fixes of the three independent reviews are in `b386446`, `fa7305f`, `b0e686e`, `9285bb6`, `d92a61b`, `c98de95`, `1561566`, `df358db` and `5b73d16`; a checklist for manual reruns and the comparison tool are in `7b86239`. Version 2: three generations and the masses and mixing (`94621af`), the neutrino flavour conversion (`a0d750a`), the electroweak sector of three generations (`cf824d5`), the removal of the tip Z portal and the cosmology with three massive neutrinos (`53c5139`), the dark matter of both halves, the twelve-fermion tables and the three-generation tip and gauge checks (`b755990`), the corrected 1-3 resonance (`b999f57`), the documents MODEL (`9a5b92b`), FIELD_EQUATIONS (`79d9d31`), SOLUTION (`a3dcc6a`), ANALYSIS (`df6d94a`) and PROVENANCE (`4a6421d`), and the fixes of the independent review of version 2 (item 22; `ecca926`). Version 3: the thermal history, the coupled solution, I13, the metric series and σ_8 (`15c2aab`), I13 on the baryon-number current (`3fa1b49`), the documents MODEL (`765f0b8`), FIELD_EQUATIONS (`41e921a`), SOLUTION (`df23013`), ANALYSIS (`19d2c06`) and PROVENANCE (`726997f`), and the fixes of the independent review of version 3 with the record of the fresh-clone verification of `726997f` (item 24; the commit that carries this text).